Lecture 06: Relational Data Model

ITCS216 Fundamentals of Database Systems

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Lesson Learning Outcome

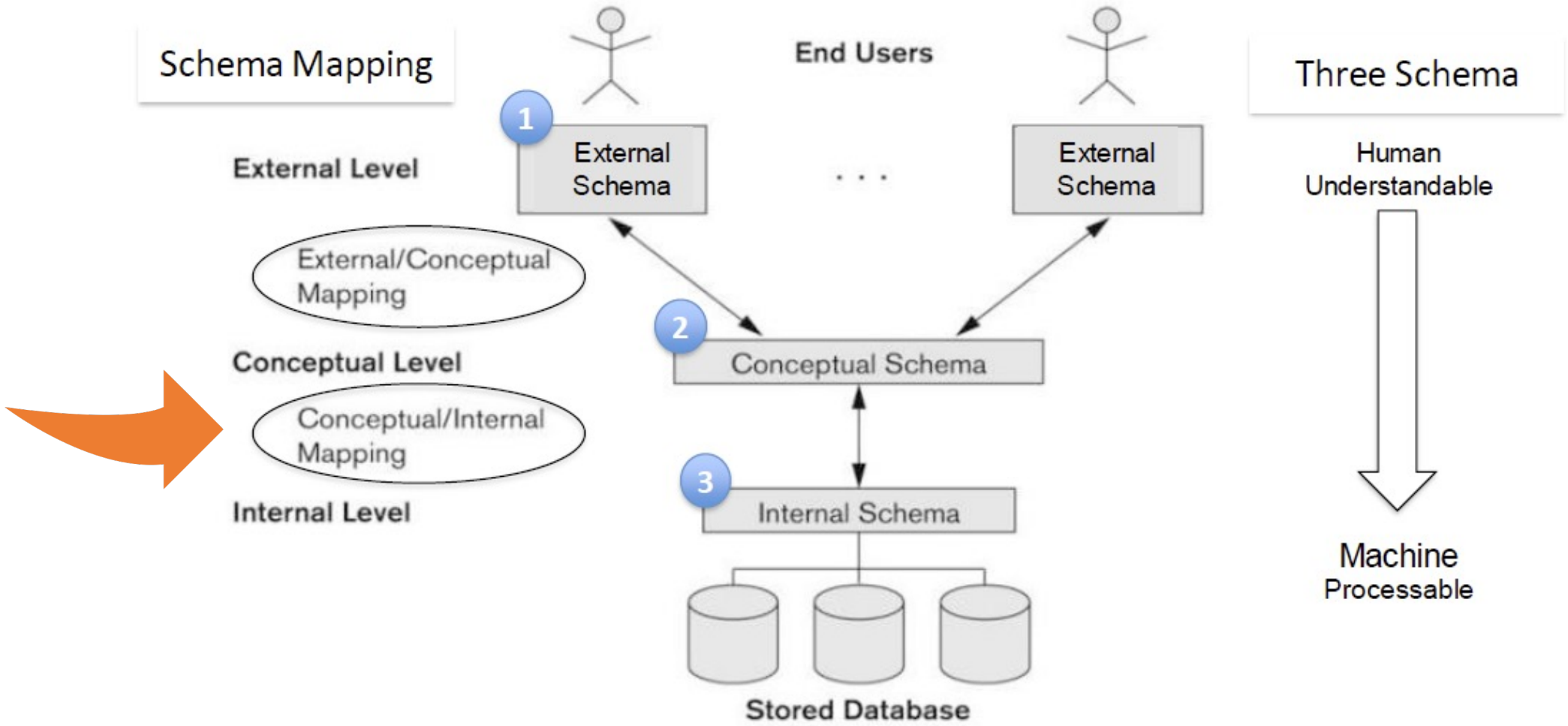
After this lesson, students are able to:

- Understand the concept of relational data model
- Differentiate between ERD and the relational data model
- Create a relational data model from the given ERD/EERD (ERD Transformation)

Database Concept

Recommend

Three Schema Architecture



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Relational Model

(Relational Schema)

Relational Data Model

- Relational Data Model offers a logical view of data rather than physical view
- Relations are implemented as tables in a relational DBMS
  - Structural and data independence
- Basic components of Relations
  - Rows or tuples
  - Columns or attributes

Relation = Table
Tuple = Row
Attribute = Column
Domain = allowable values

ERD vs. Relational Data Model

ER Model

- Abstract concept of the database
- Deals with entities and their relationships
- Describes the data with entity set, relationship set and attributes
- Easily understand relationship among data
- Contains cardinality as a constraint

Relational Model

- Deals with tables and relation between the data of the tables
- Describes the data with the tuples, attributes, and domain of attributes
- Less easy to derive relationship
- Cardinality is not directly shown in the relational schema (only present PK/FK)
- Formats and relations in a way a database could understand (DB implementation)

Table and Table Characteristics

Logical view of relational database is based on relation -> Relation thought of *tables*

Table: two-dimensional structure composed of rows and columns

- Persistent representation of logical relation

Contains group of related entities (entity set)

TABLE 3.1

CHARACTERISTICS OF A RELATIONAL TABLE

1	A table is perceived as a two-dimensional structure composed of rows and columns.
2	Each table row (**tuple**) represents a single entity occurrence within the entity set.
3	Each table column represents an attribute, and each column has a distinct name.
4	Each intersection of a row and column represents a single data value.
5	All values in a column must conform to the same data format.
6	Each column has a specific range of values known as the attribute **domain**.
7	The order of the rows and columns is immaterial to the DBMS.
8	Each table must have an attribute or combination of attributes that uniquely identifies each row.

Primary Key

Primary Key

- **Primary Key (PK)** - Attribute or combination of attributes that uniquely identifies any given row in a table, ensuring data integrity.
- Characteristics:
  - Must be unique for every row
  - Cannot contain NULL values
  - A table can have only one primary key
- Source:
  - The primary key is often the ***natural key*** of the table derived from existing data (e.g., person's email address), but it can also be a ***surrogate key***.

Surrogate Key

- **Surrogate key** is an artificial key generated by the database that has no business meaning. It is purely for database purpose – to uniquely identify each record.
- Source:
  - An artificially generated key, such as an auto-incrementing number or UUID
- Characteristics:
  - Has no business meaning or real-world value
  - Often used as PK when no suitable natural key exists or when a natural key is too long, complex or subject to change

Natural Key, Surrogate Key, PK

- For example, Customer data may have
- **Natural Key**: A customer's email address
- **Surrogate Key**: A unique customer ID number

- In this example, email address can uniquely identify a row of customer in a table. However, a customer may change his/her email address. This can cause problem in updating PK in the database. To avoid this issue, a surrogate key can be used as PK instead of the natural key. It is typically not subject to change.

Foreign Key

- A **foreign key** is an attribute or set of attributes that references the **primary key of another relation**.

Employee

SSN	FNAME	LNAME	GENDER	DEPT_NO (FK)
999887777	Alicia	Wallace	F	4
987654321	Alicia	Zelaya	F	5
666884444	Ramesh	Narayan	M	5
453453453	Bob	Wallace	M	1

Department

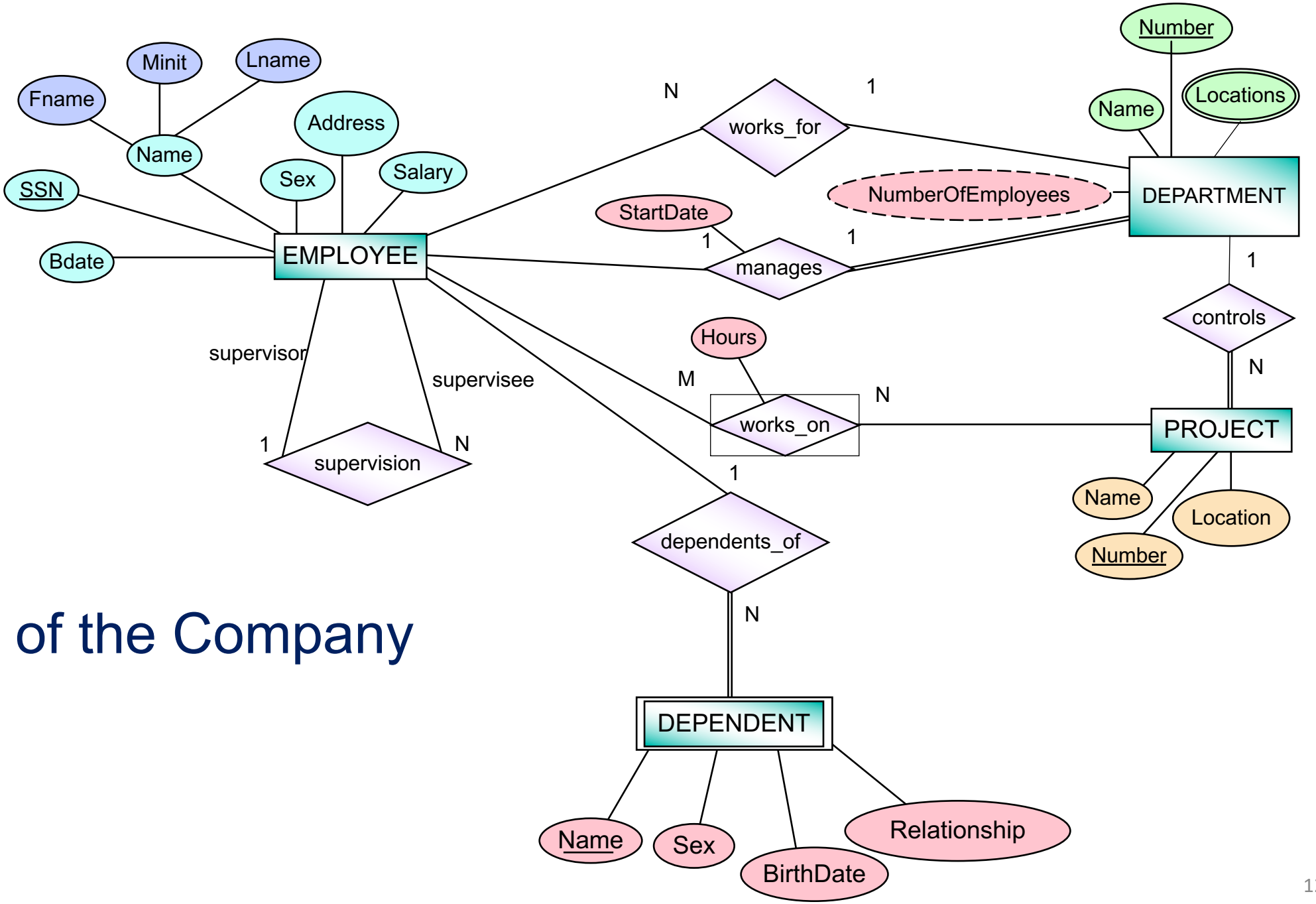
DEPT_NO (PK)	NAME	LOCATION
1	Admin	Los Angeles
2	Human Resource	Los Angeles
3	Marketing	Los Angeles
4	Finance	Los Angeles
5	IT	San Francisco

FK *may be NULL if the relationship is optional*.

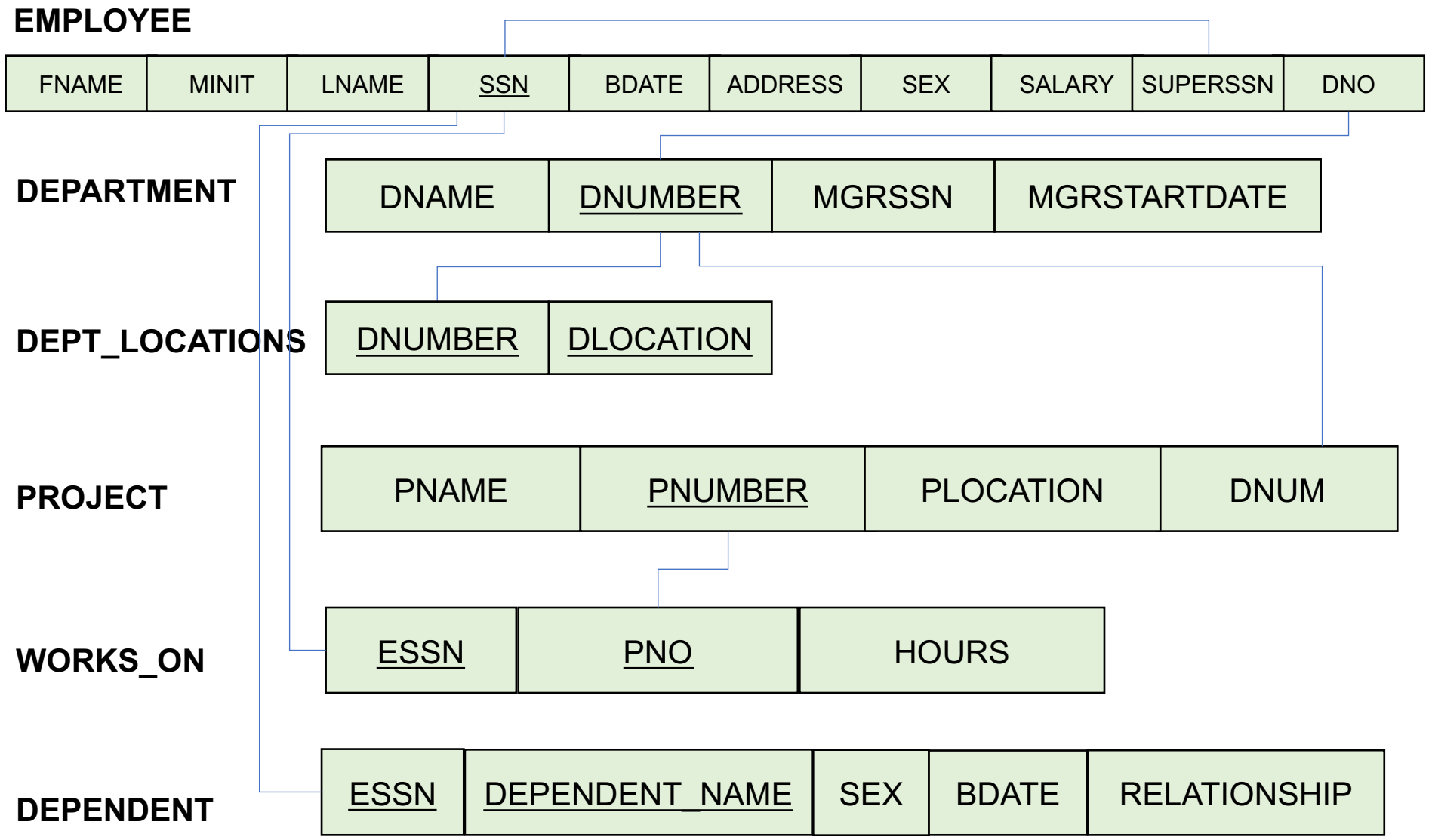
In short: “PK identifies my row; FK connects my row to another table.”

****IMPORTANT** you must NOT include FK in your ERD/EERD**

ERD of the Company



Example: Relational Data Model



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ERD/EERD Transformation

Mapping ERD to Relational Model

ERD Transformation

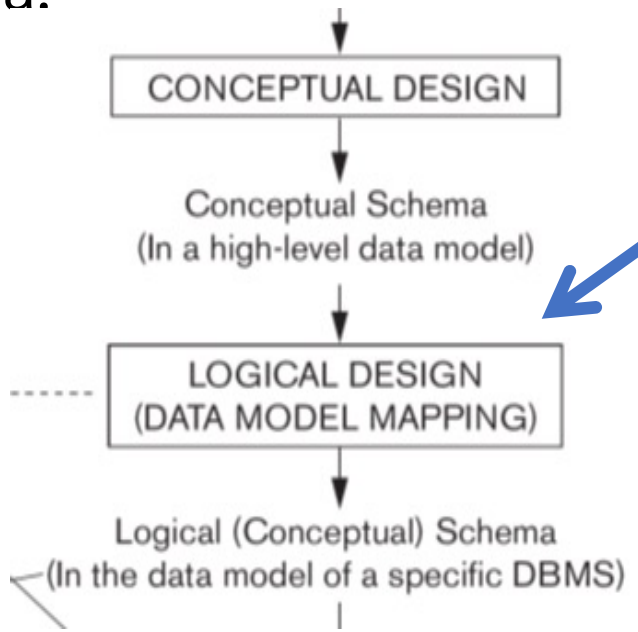
An EDR/EERD represents

a **conceptual** schema, whereas

relational model is a **logical** schema.

So we need to "transform"

ERD/EERD to relational model.

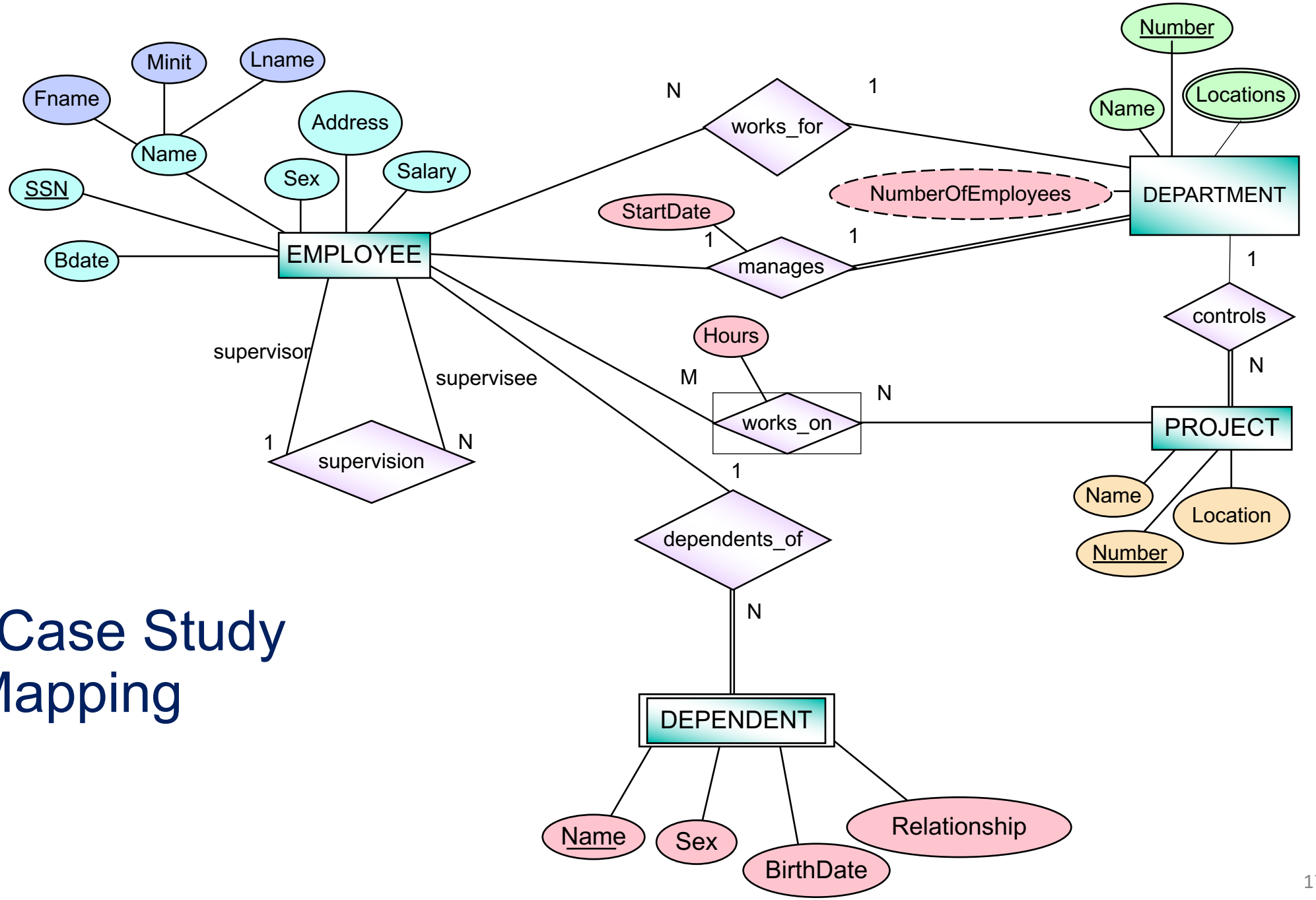


Mapping algorithm

8 Steps to Transform ERD to Relational Model

1. Transforming regular entity types
2. Transforming weak entity types
3. Transforming binary 1:1 relationship types
4. Transforming binary 1:N relationship types
5. Transforming binary M:N relationship types
6. Transforming multi-valued attributes *A* in relation *K*
7. Transforming *n-ary* relationship types
8. Transforming specialization or generalization

Our Case Study for Mapping



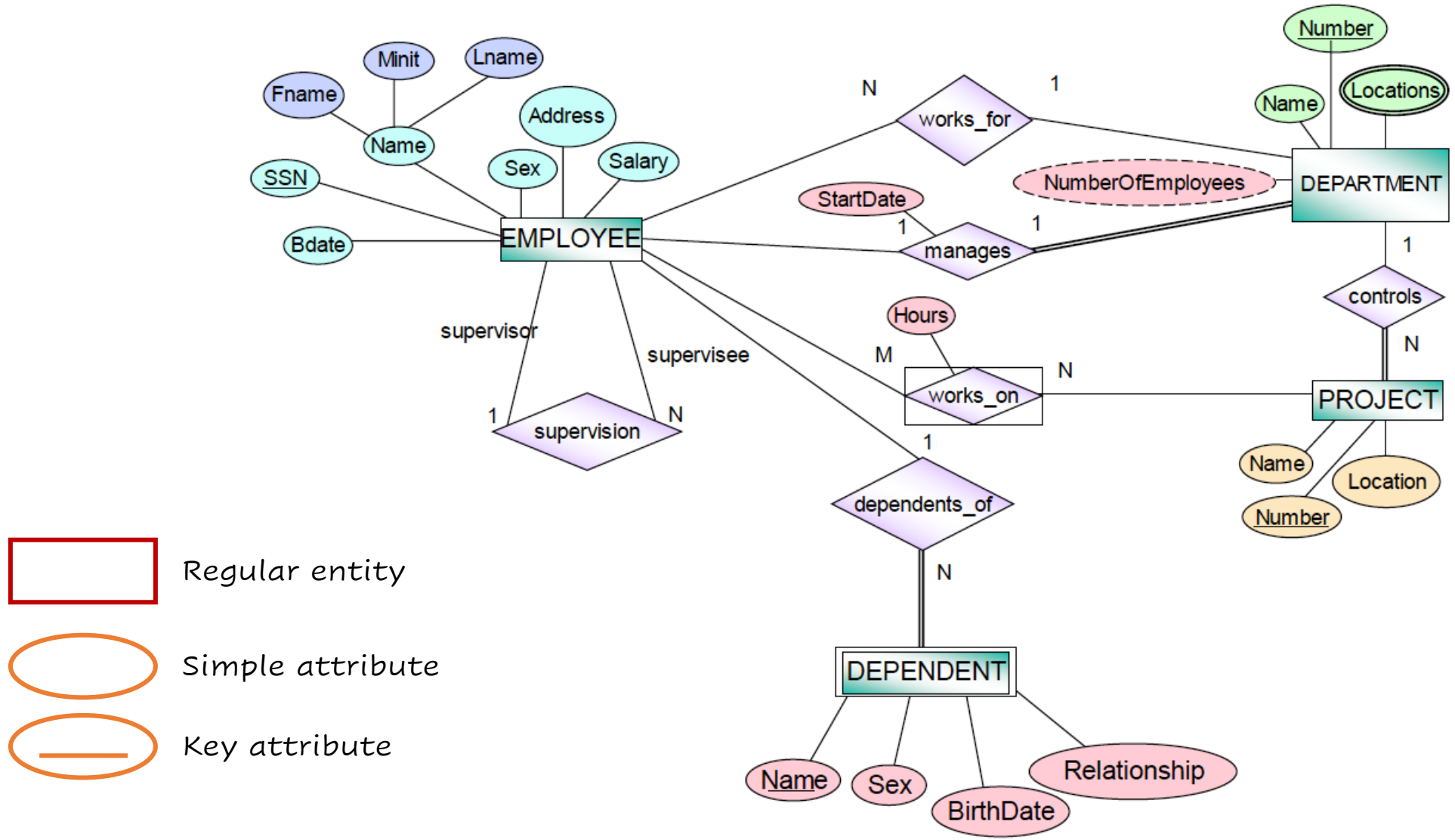
ER-to-Relational Mapping Algorithm

Step 1: Create a **relation R** for each regular **entity type E**

- Include only the **simple attributes** (of the composite attributes)
- Declare each **primary key**
  - (if necessary combines a set of attributes to form a PK)

Primary key is the key attribute(s) that is used to uniquely identify a data row

“Step 1: Convert regular entities and simple attributes”



Step 1: Regular Entity (cont.)

Relation: Employee

EMPLOYEE

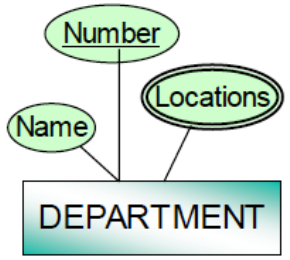
SSN	FNAME	MINIT	LNAME	BDATE	ADDRESS	SEX	SALARY

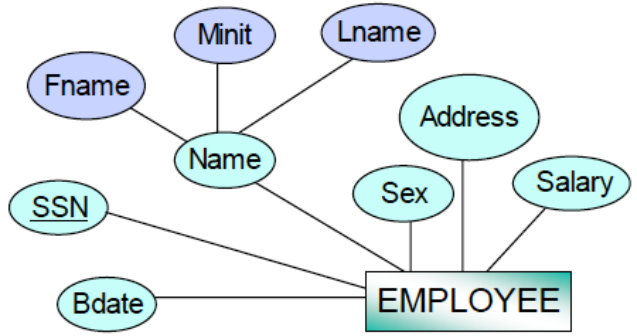
Or you can write:

EMPLOYEE(SSN, FNAME, MINIT, LNAME, BDATE, ADDRESS, SEX, SALARY)

DEPARTMENT

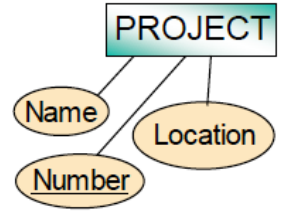
DNAME	**DNUMBER**





PROJECT

PNUMBER	PNAME	PLOCATION



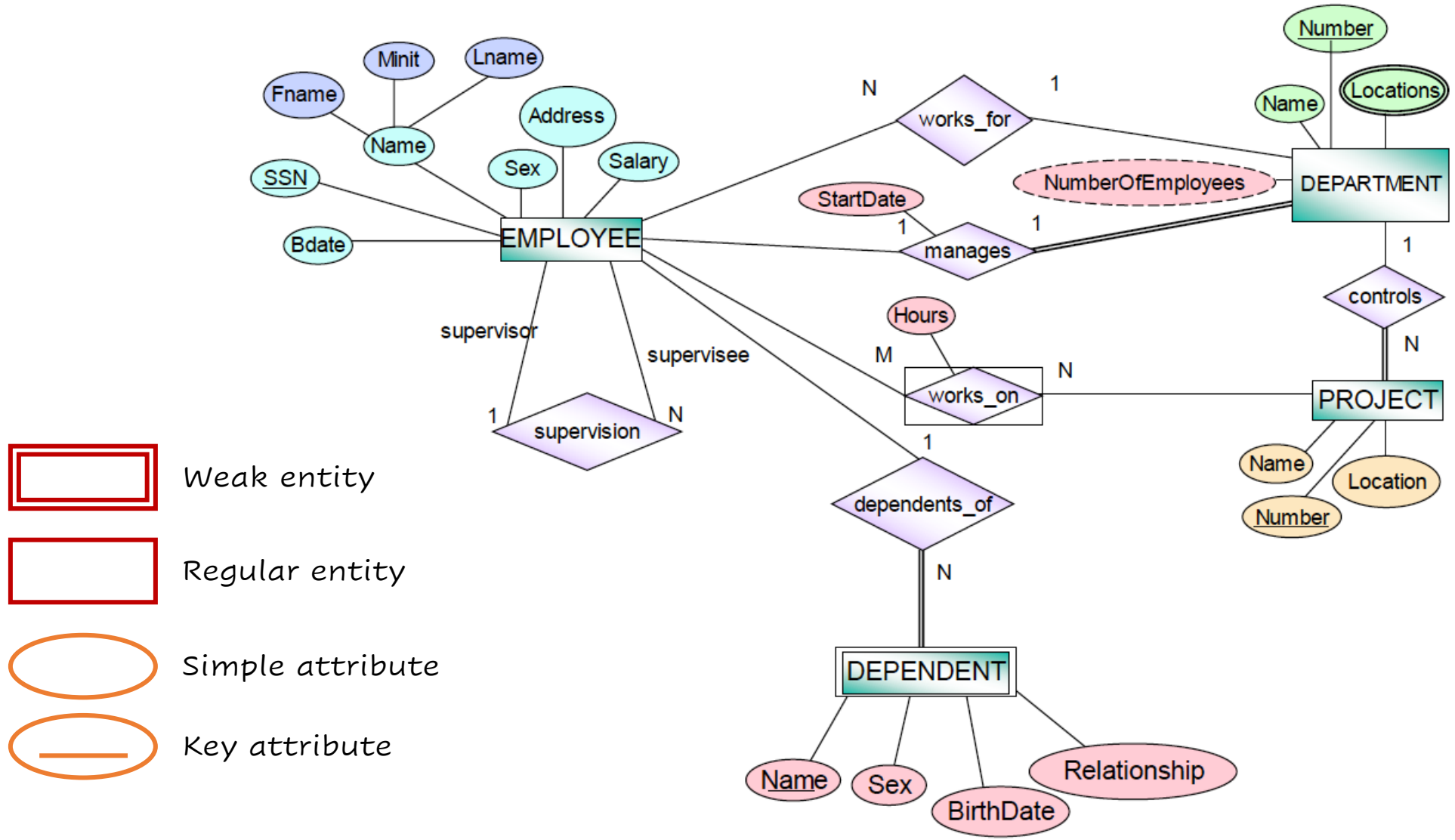
Step 2 : Create a relation R for each weak entity type W

- Include the attributes
- Include a FK of PK of its owner (strong) entity type.
- The PK of R is the combination of PK of the owner(s) and the partial key of the weak entity W.

Primary key (PK) is the key attribute(s) that is used to uniquely identify a data row

Foreign key (FK) is the attribute(s) that match the Primary key of another relation (i.e., take PK of another relation to be one of its own attributes)

“Step 2: Convert weak entities”



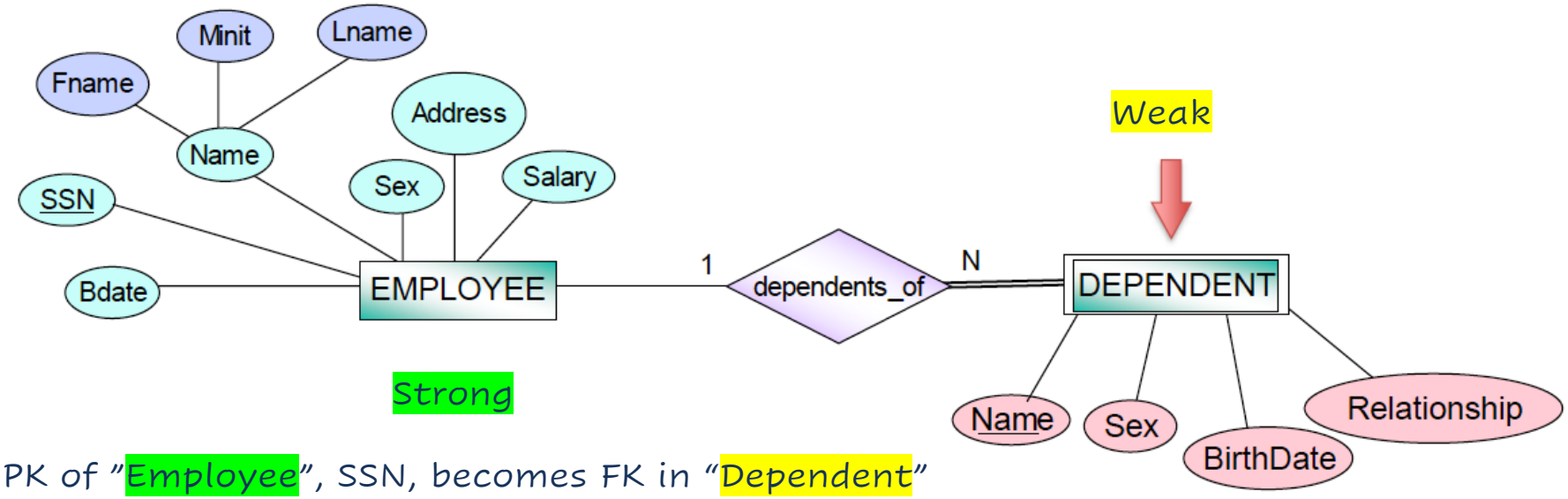
SSN	FNAME	MINIT	LNAME	BDATE	ADDRESS	SEX	SALARY

Step 2: Weak Entity

DEPENDENT

ESSN	DEPENDENT_NAME	SEX	BDATE	RELATIONSHIP

important
ESSN is a foreign key that refers to the SSN (primary) key in the relation EMPLOYEE.
ESSN **must** be a part of a combination of primary key in the relation DEPENDENT



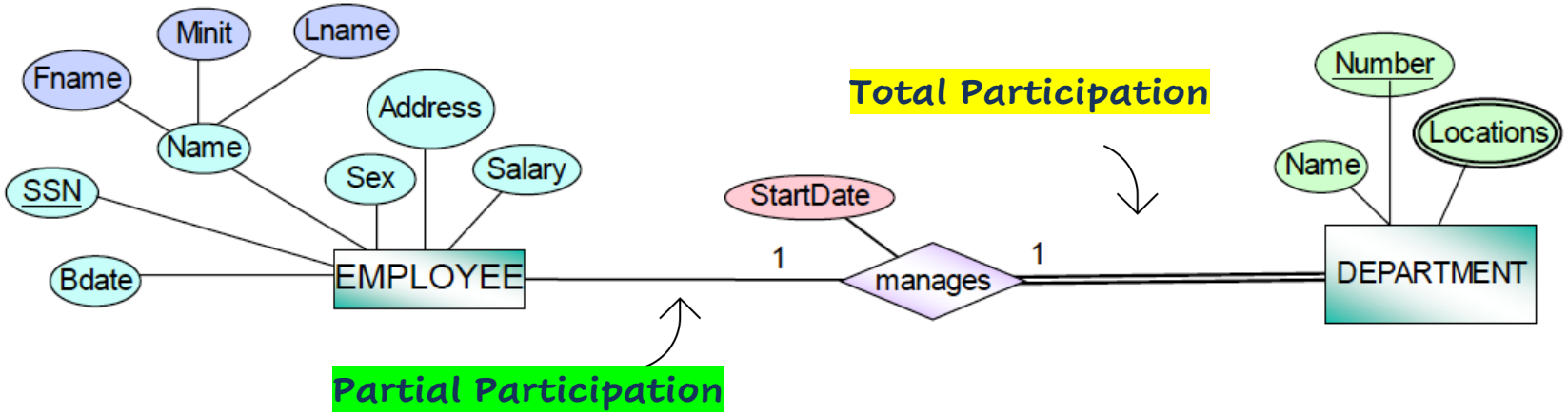
The PK of "Employee", SSN, becomes FK in "Dependent" and forms a composite PK together with the partial PK DEPEDENT_NAME (existing before).

Step 3 : Binary 1: 1 → Relationship Type R

- Identify two entities **E1** and **E2** participating in **R**
- Choose one with total participation in R say **E1**
- Include a FK in **E1** obtained from the PK of **E2**
- Include the attributes of relationship to **E1**

1:1 → Place the FK on the total-participation side

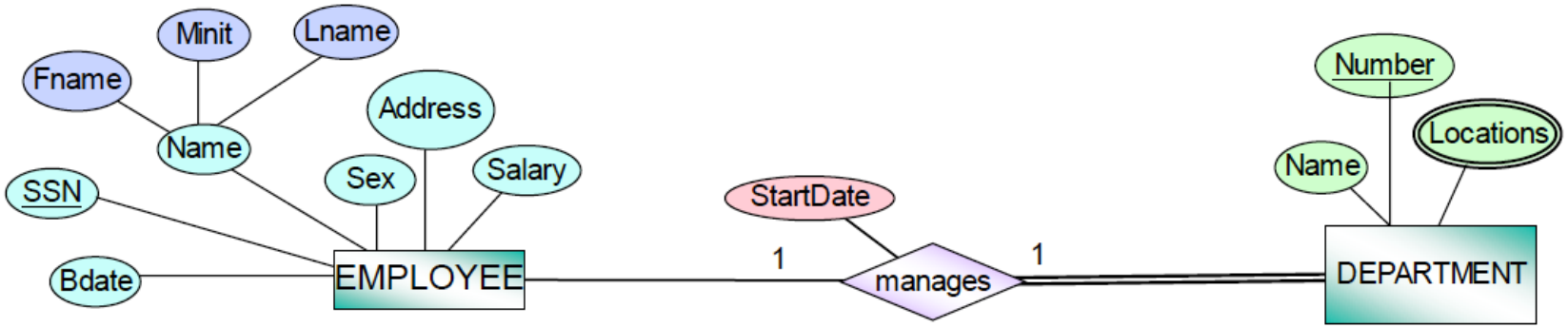
Step 3: Binary 1:1 → Relationship Type R



From Step 3, Because "DEPARTMENT" has total participation in "manages" relationship, we include EMPLOYEE.SSN as FK in "DEPARTMENT".

FK can be renamed as needed → MGRSSN as Manager's SSN

Step 3: Binary 1:1 → Relationship Type R



DEPARTMENT

DNUMBER	DNAME	**MGRSSN**	**MGRSTARTDATE**

important
MGRSSN is a foreign key that refers to the SSN (primary) key in the relation EMPLOYEE.
MGRSSN is **not** a part of primary key in relation DEPARTMENT

EMPLOYEE

SSN	FNAME	MINIT	LNAME	BDATE	ADDRESS	SEX	SALARY

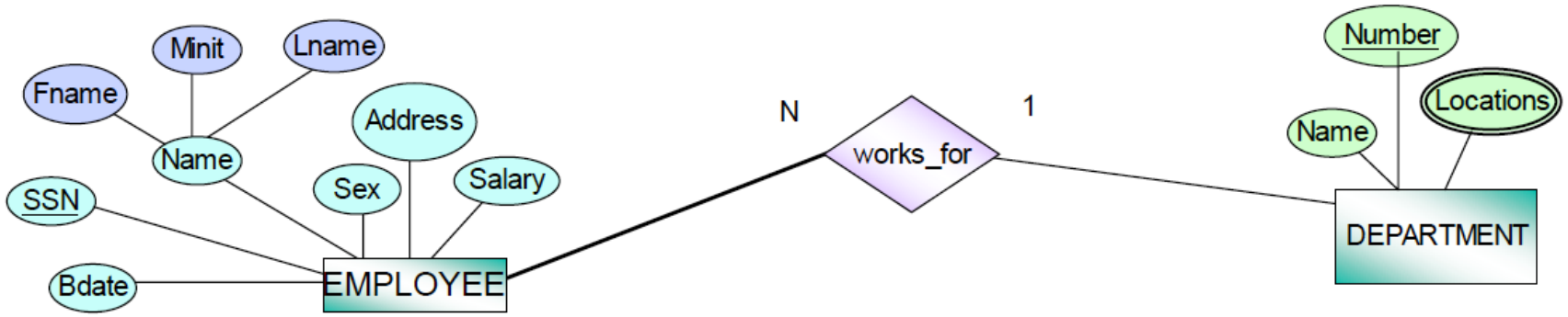
(note. Relation EMPLOYEE is still the same as shown in step 1)

Step 4 : Binary 1:N → Relationship Type R

- Identify the entity **E1** that represent the N-side of the relationship.
- Include a FK in **E1** obtained from the PK of the entity of the other side.
- Include the attributes of relationship to **E1.**

1:M → PK from the 1-side becomes FK on the N-side

“Step 4: Convert 1-M Relationship”

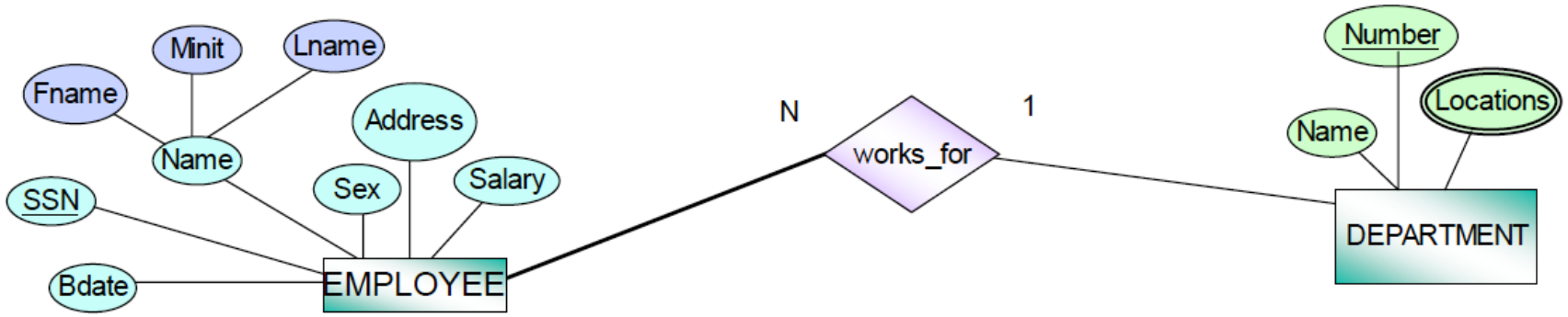


N side = Employee

1 side = Department

From Step 4, we include FK at the “Employee” obtained from the PK of “Department”

"Step 4: Convert 1-M Relationship" – Cont.



From Step 4, we include FK at the N-side "Employee" obtained from the PK of 1-side "Department"

Example: At Employee, we include "DNO" as the department number from "DNUMBER" of Department

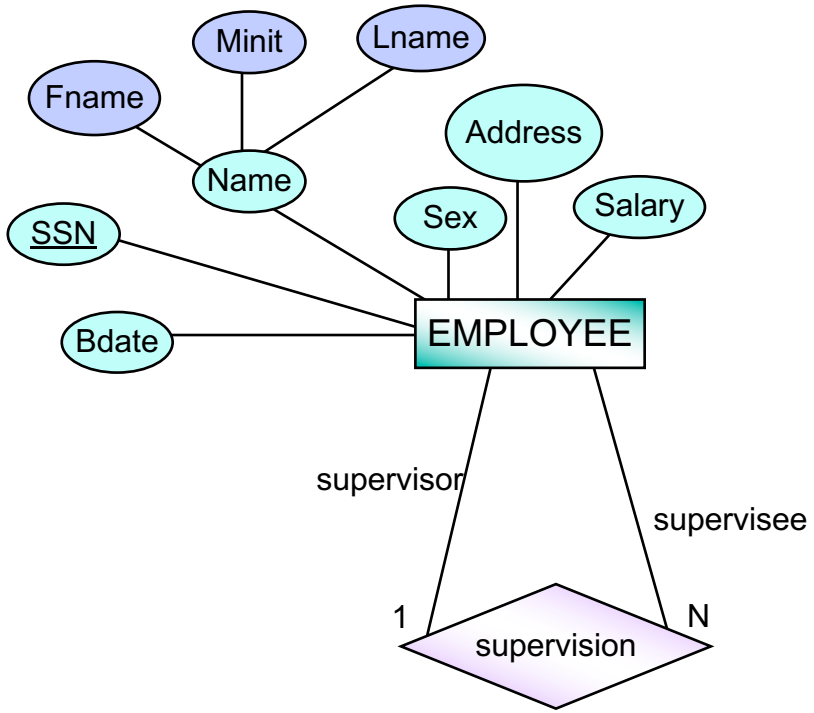
DNUMBER	DNAME	MGRSSN	MGRSTARTDATE

DEPARTMENT

EMPLOYEE

SSN	FNAME	MINIT	LNAME	BDATE	ADDRESS	SEX	SALARY	**DNO**

"Step 4: Convert 1-M Relationship" – Cont.



From Step 4, we include FK at the N-side "Employee" obtained from the PK of 1-side "Employee"

Example: At Employee, we include "SUPERSSN" as the supervisor SSN from "SSN" of Employee itself.

EMPLOYEE

SSN	FNAME	MINIT	LNAME	BDATE	ADDRESS	SEX	SALARY	DNO	SUPERSSN

"Step 4: Convert 1-M Relationship" – Cont.

From Step 4, we include FK at the N-side "Project" obtained from the PK of 1-side "Department"

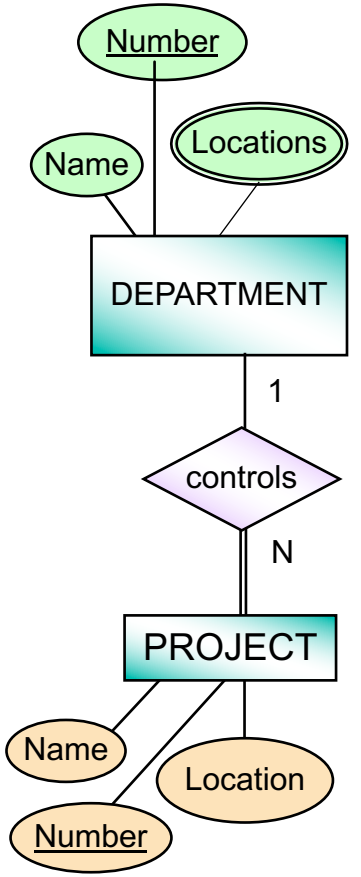
Example: At project, we include "DNUM" as the department number from "DNUMBER" of department

DEPARTMENT

DNUMBER	DNAME	MGRSSN	MGRSTARTDATE

PROJECT

PNUMBER	PNAME	PLOCATION	DNUM



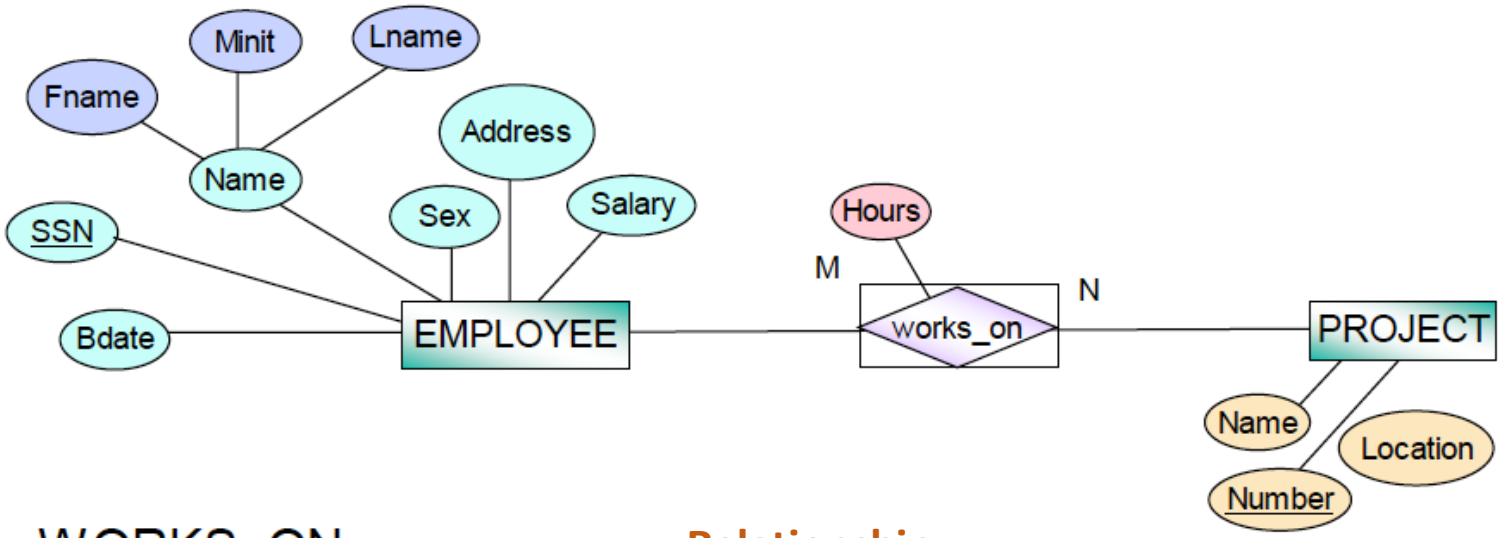
Note that we do not worry about *total or partial participation* in this step. ONLY focus on connectivity constratin (1:M)

Step 5 : Binary M:N → Relationship Type R

- Create a new relation **E3** to represent R
- Include the FK in **E3** obtained from the PK of the two entities then use the combination of FK to form the PK of **E3**
- Include the attributes of relationship to **E3**

M:N → Create a new relation

Step 5: Binary M:N → Relationship Type R



WORKS_ON — Relationship

PK,FK ESSN	PK,FK PNO	Attribute HOURS

Create a new relation to represent the M:N relationship

important
ESSN is a foreign key that refers to the SSN (primary) key in the relation EMPLOYEE.
PNO is a foreign key that refers to the Number (primary) key in the relation PROJECT.
Both ESSN and PNO together must form a primary key in the relation WORKS_ON.

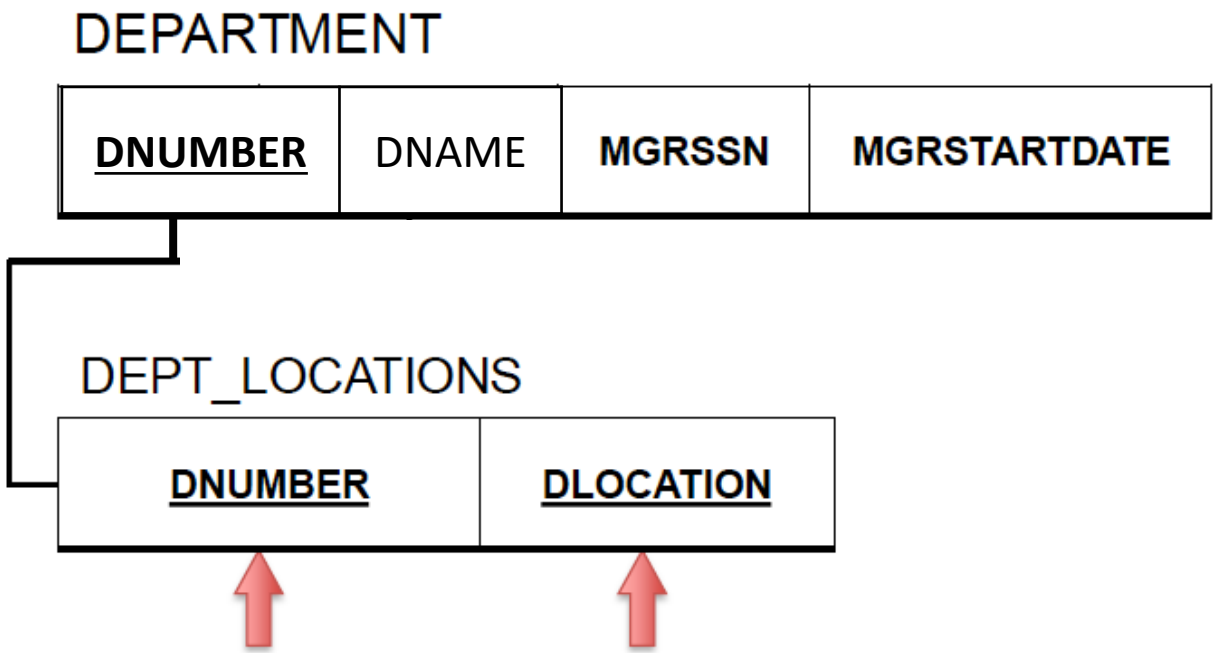
Step 6 : Multi-Valued Attribute

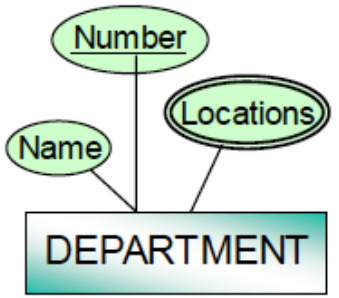
For each multivalued attribute **A**

- Create a new relation R for that attribute
- Include an attribute corresponding to **A** plus a PK of the relation that **A** belongs to
- The PK of R is the combination of **A** and **K**

Create a new relation

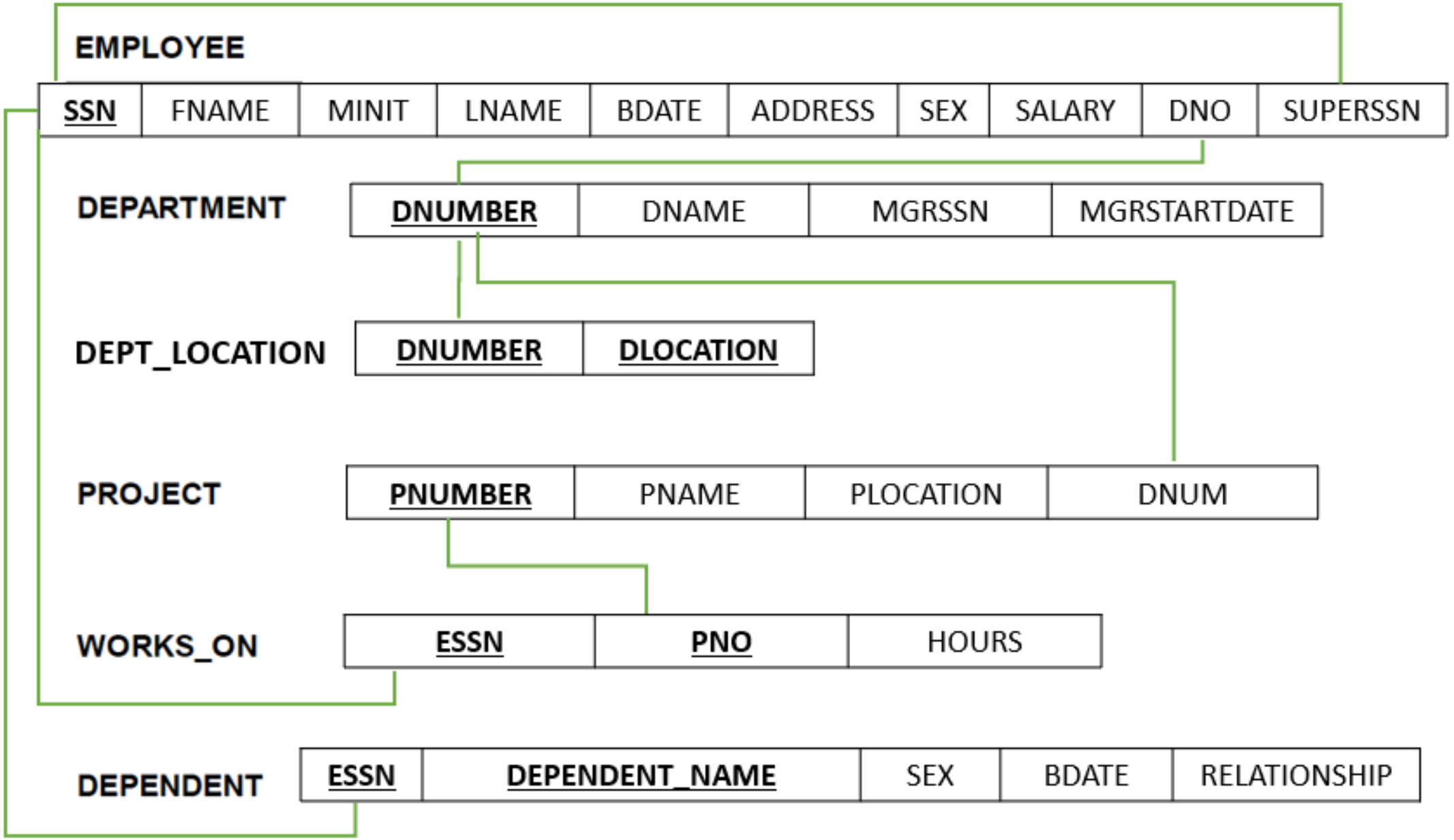
Step 6: Multi-Valued Attribute





Create a new relation: multivalued attribute, PK of its owner as na FK, use both as the composite PK

The Company Relational Schema



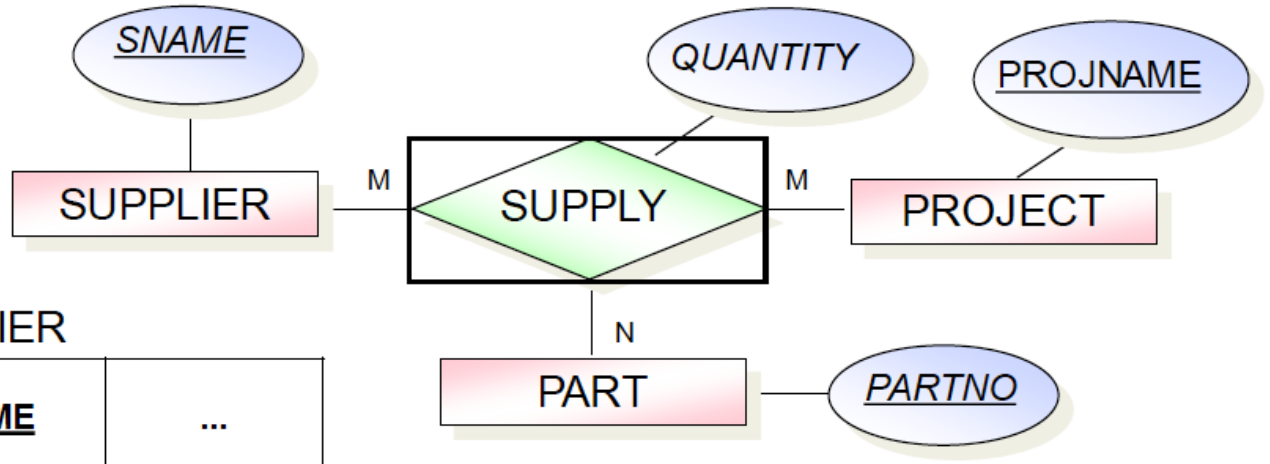
Step 7 : N-ary Relationship

For each n-ary relationship type R (n>2)

- Create a new relation **E** to represent R
- Include the FK in S obtained from the PK of the participating entities
- Include the attributes of the n-ary relationship to **E**

Create a new relation and include the PKs of all participating entities as foreign keys

Step 7: N-ary Relationship



SUPPLIER

SNAME	...

Each supplier supplies several parts for any one of many projects. Each project has many parts from any one of many suppliers.

PROJECT

PROJNAME	...

PART

PARTNO	...

SUPPLY

SNAME	PROJNAME	PARTNO	QUANTITY

“Step 8: Convert Specialization / Generalization (Specialization hierarchy in EERD)”

4 methods to convert; each has their limitations except 8A

Specialization constraint	8A	8B	8C	8D
Total – Disjoint	✓	✓	✓	✓
Total – Overlapping	✓	✓	✗	✓
Partial – Disjoint	✓	✗	✓	✓
Partial – Overlapping	✓	✗	✗	✓

8A — Supertype relation + one relation for each subtype
8B — Subtype relations only
8C — One relation + one type/discriminator attribute
8D — One relation + multiple type/Boolean attributes

Transforming EERD to Relational Model

Step 8 : Transforming Specialization or Generalization

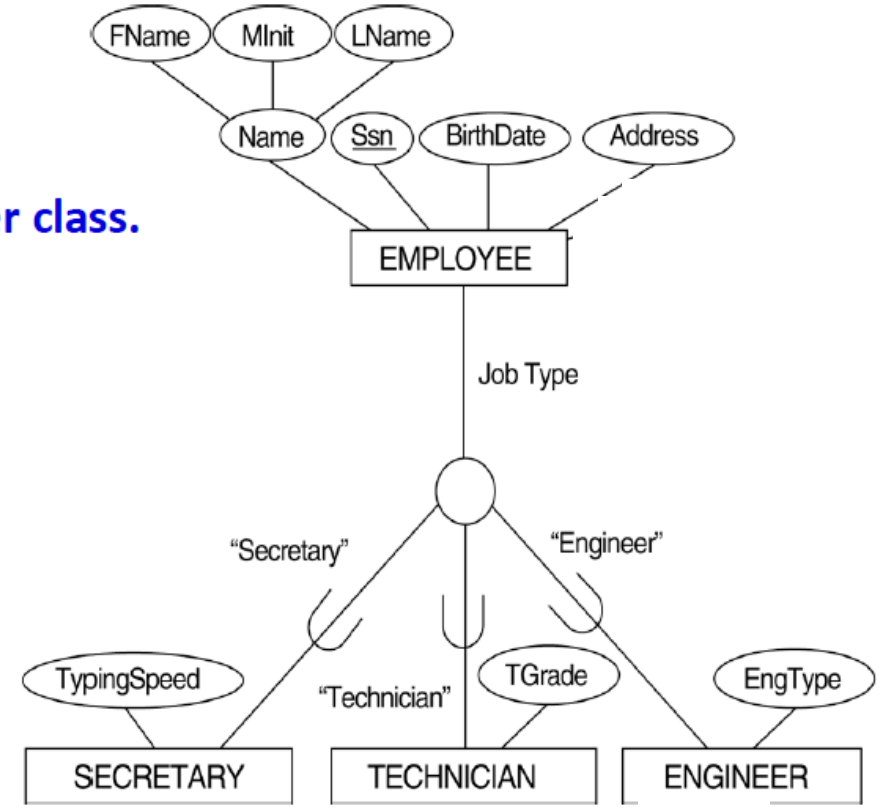
Option 8A: Multiple relations, Super class and subclasses.

- Create a relation for the super class, including the super class attributes.
- Create a relation for each subclass, which includes the primary key of the super class (which acts as the foreign key) and the attributes of the subclass specialization.
- This works for any specialization (partial, total, disjoint, overlapping)

In Method 8A, inherits only the PK from Super Class

Step 8: Transforming Specialization or Generalization

Method 8A (for all cases):
Inherits only the PK from Super class.



(a)

EMPLOYEE

SSN	FName	MInit	LName	BirthDate	Address

SECRETARY

SSN	TypingSpeed

TECHNICIAN

SSN	TGrade

ENGINEER

SSN	EngType

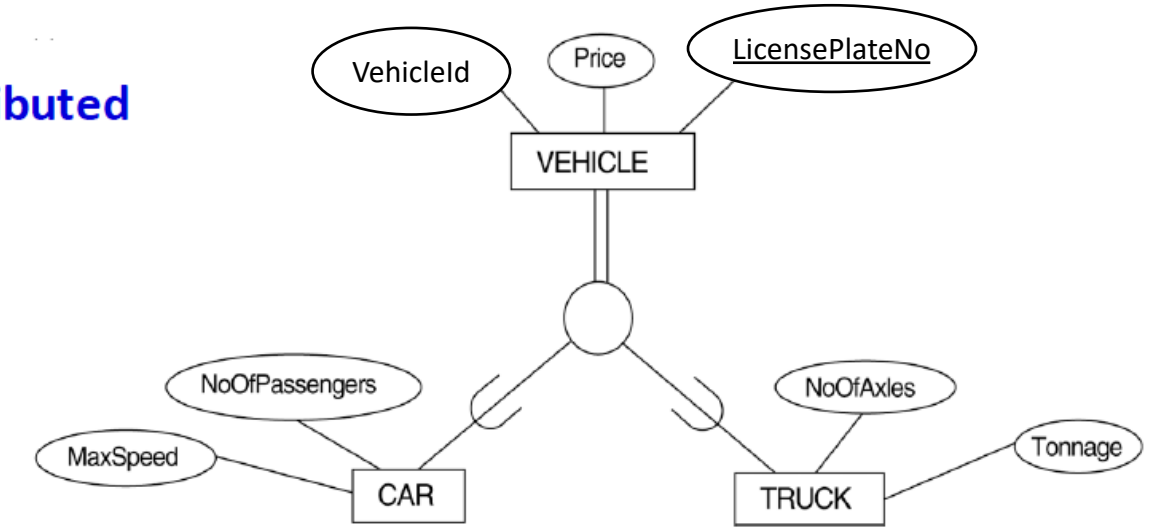
Step 8: Transforming Specialization or Generalization (cont.)

Option 8B: Multiple relations, Subclass relations only

- Create a relation for each subclass, with the attributes of both the super class and the attributes of the subclass.
- This only works for total specializations, meaning that every entity in the super class must belong to at least one subclass. Otherwise members of the super class that don't belong to a subclass will not be represented.

Step 8: Transforming Specialization or Generalization (cont.)

Method 8B (for TOTAL)
Inherits all common attributed from Super class.



(b)

CAR

LicensePlateNo	VehicleId	Price	MaxSpeed	NoOfPassengers

TRUCK

LicensePlateNo	VehicleId	Price	NoOfAxles	Tonnage

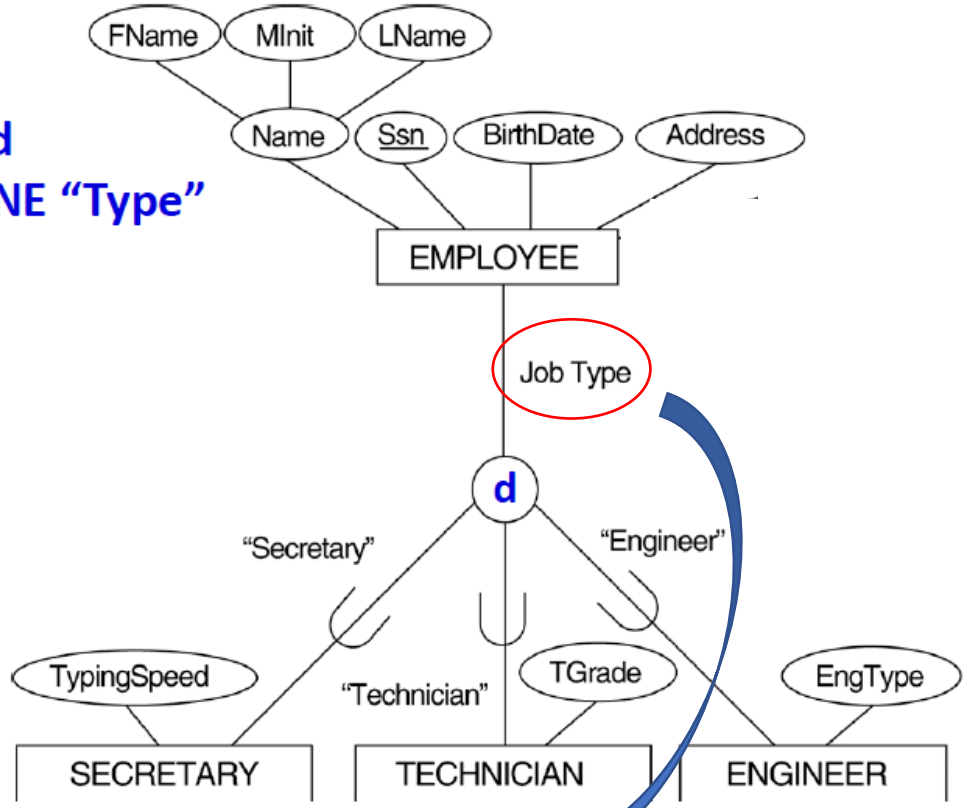
Step 8: Transforming Specialization or Generalization (cont.)

Option 8C: Single relation with one type attribute.

- Create a single relation, with all the attributes of the super class and all the attributes of a subclass.
- Include a 'Type' attribute, which is the discriminating attribute which indicates which subclass the row belongs to.
- This only works if the specialization is disjoint, meaning the super class entity cannot be a member of more than one subclass.

Step 8: Transforming Specialization or Generalization (cont.)

Method 8C (for DISJOINT), Inherits all common attributed from Super class and assign ONE "Type" attribute for sub class.



(c) EMPLOYEE

SSN	FName	MInit	LName	BirthDate	Address	JobType	TypingSpeed	TGrade	EngType

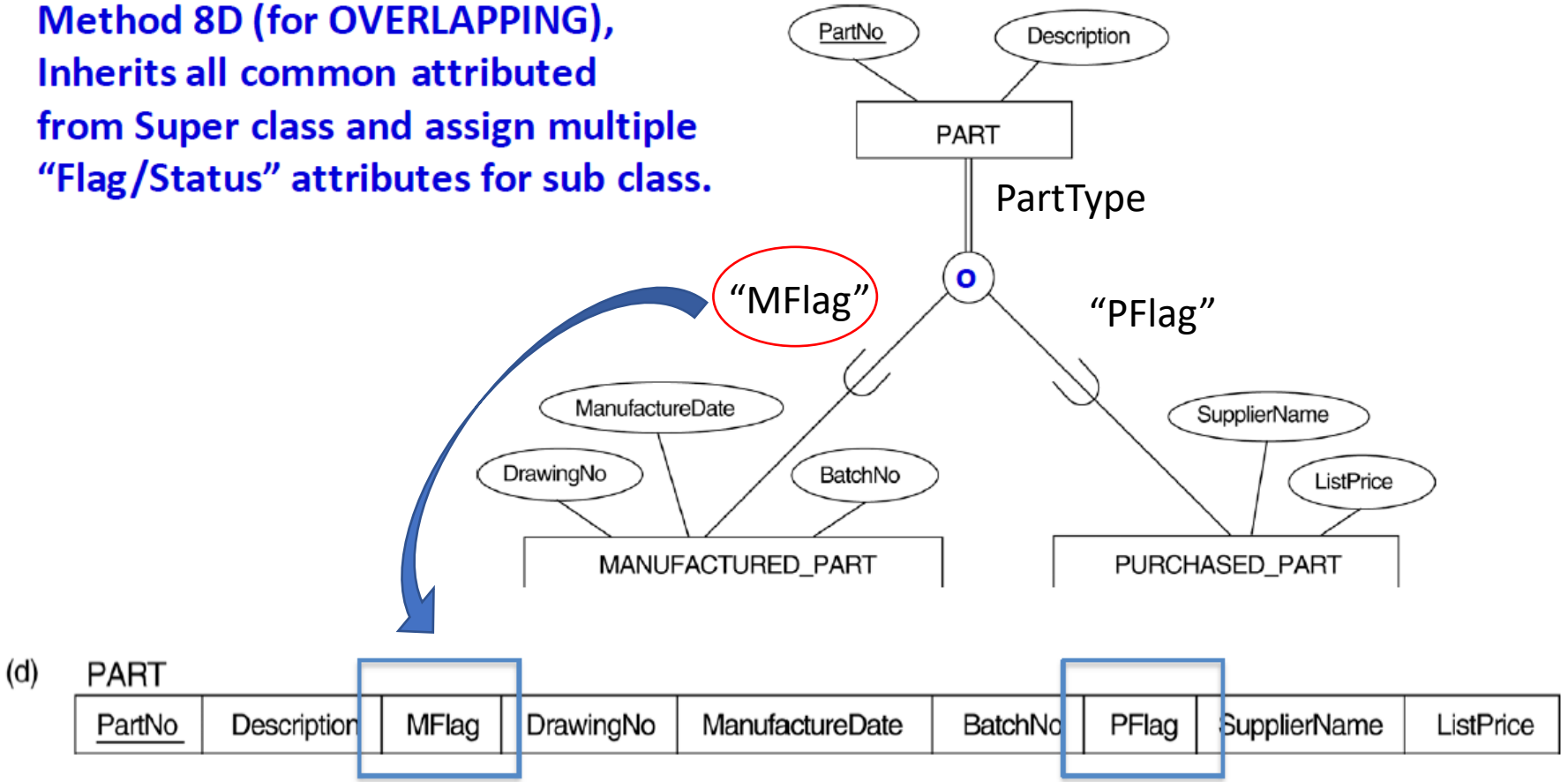
Step 8: Transforming Specialization or Generalization (cont.)

Option 8D: Single relation with multiple type attributes

- Create a single relation with all the attributes of the super class and all the attributes of the subclass.
- Include a Boolean "Type" attribute for each subclass, which indicates whether the row belongs to that subclass.
- This works with overlapping specializations, to indicate if the super class entity belongs to more than one subclass.

Step 8: Transforming Specialization or Generalization (cont.)

Method 8D (for OVERLAPPING), Inherits all common attributed from Super class and assign multiple "Flag/Status" attributes for sub class.



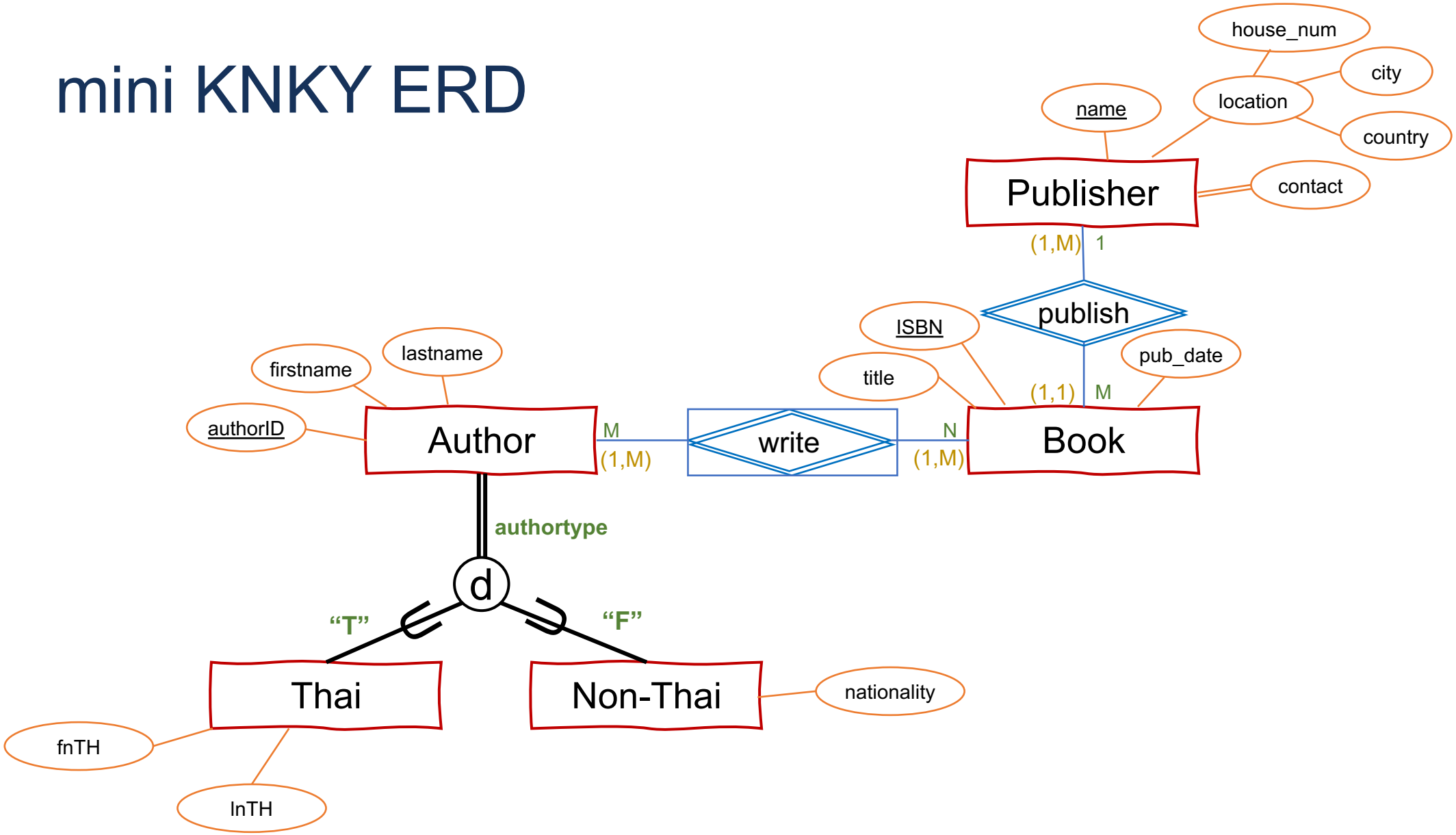
8 Steps to Transform ERD to Relational Model

1. Transforming regular entity types
2. Transforming weak entities
3. Transforming binary 1:1 relationship types
4. Transforming binary 1:N relationship types
5. Transforming binary M:N relationship types
6. Transforming multi-valued attribute *A* in relation *K*
7. Transforming *n-ary* relationship types
8. Transforming specialization or generalization

Each step can be skipped if your ERD/EERD does not contain those criteria

Case Study: mini KNKY

mini KNKY ERD



8 Steps to Transform ERD to Relational Model

1. Transforming regular entity types
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8. Transforming specialization or generalization

Each step can be skipped if your ERD/EERD does not contain those criteria

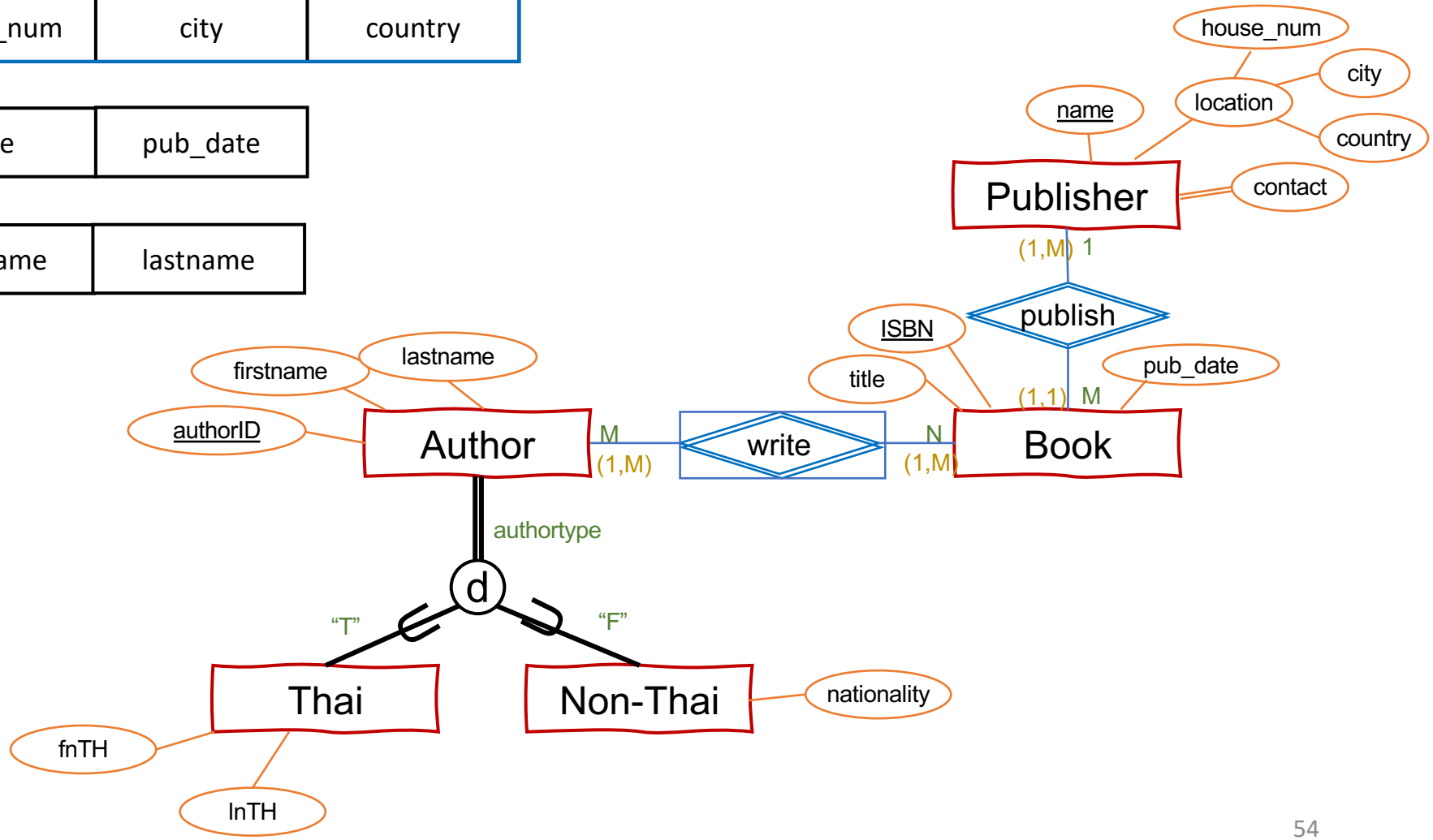
Step 1: Regular Entity

location

Publisher	name	house_num	city	country

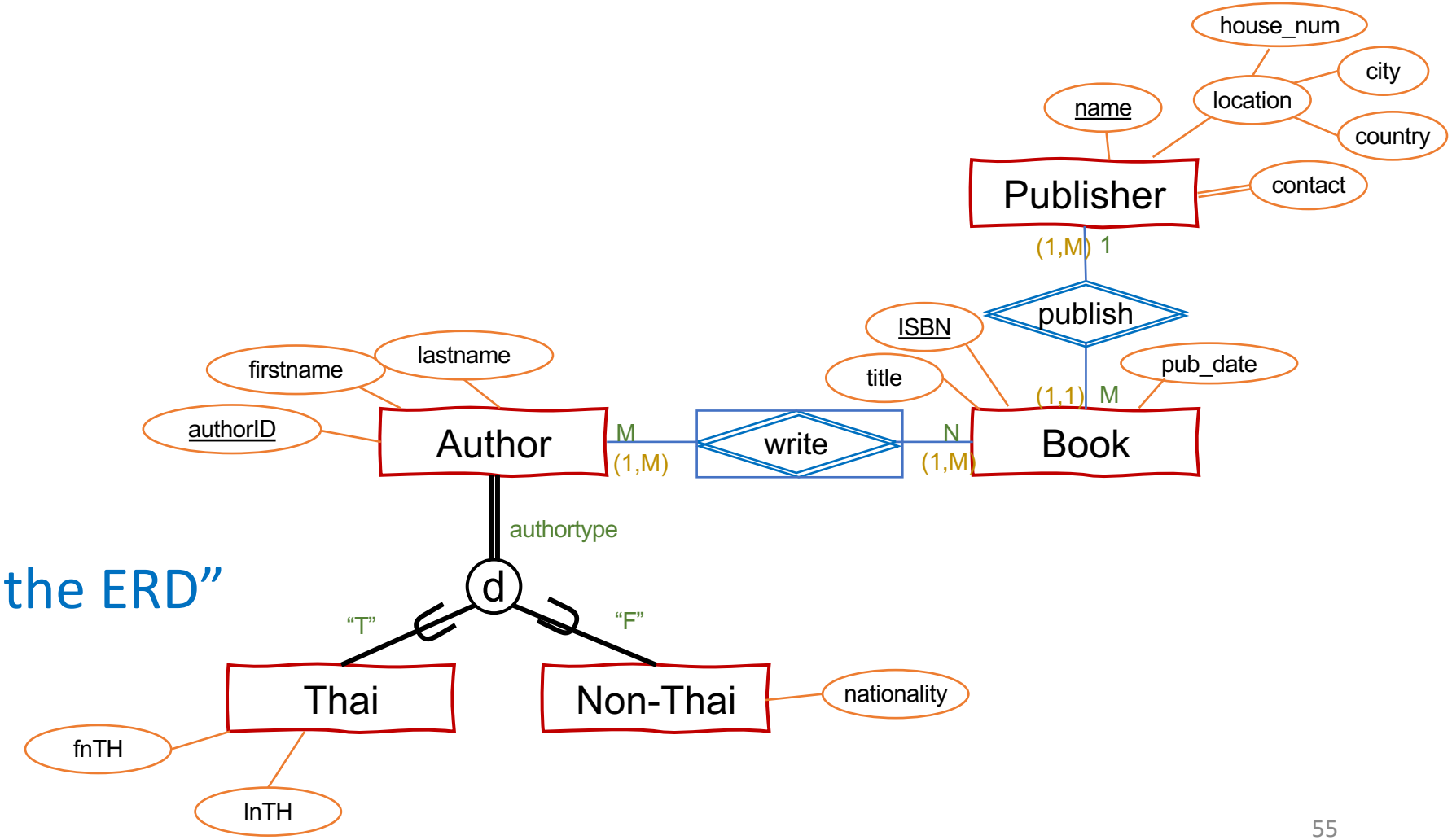
Book	ISBN	title	pub_date

Author	authorID	firstname	lastname



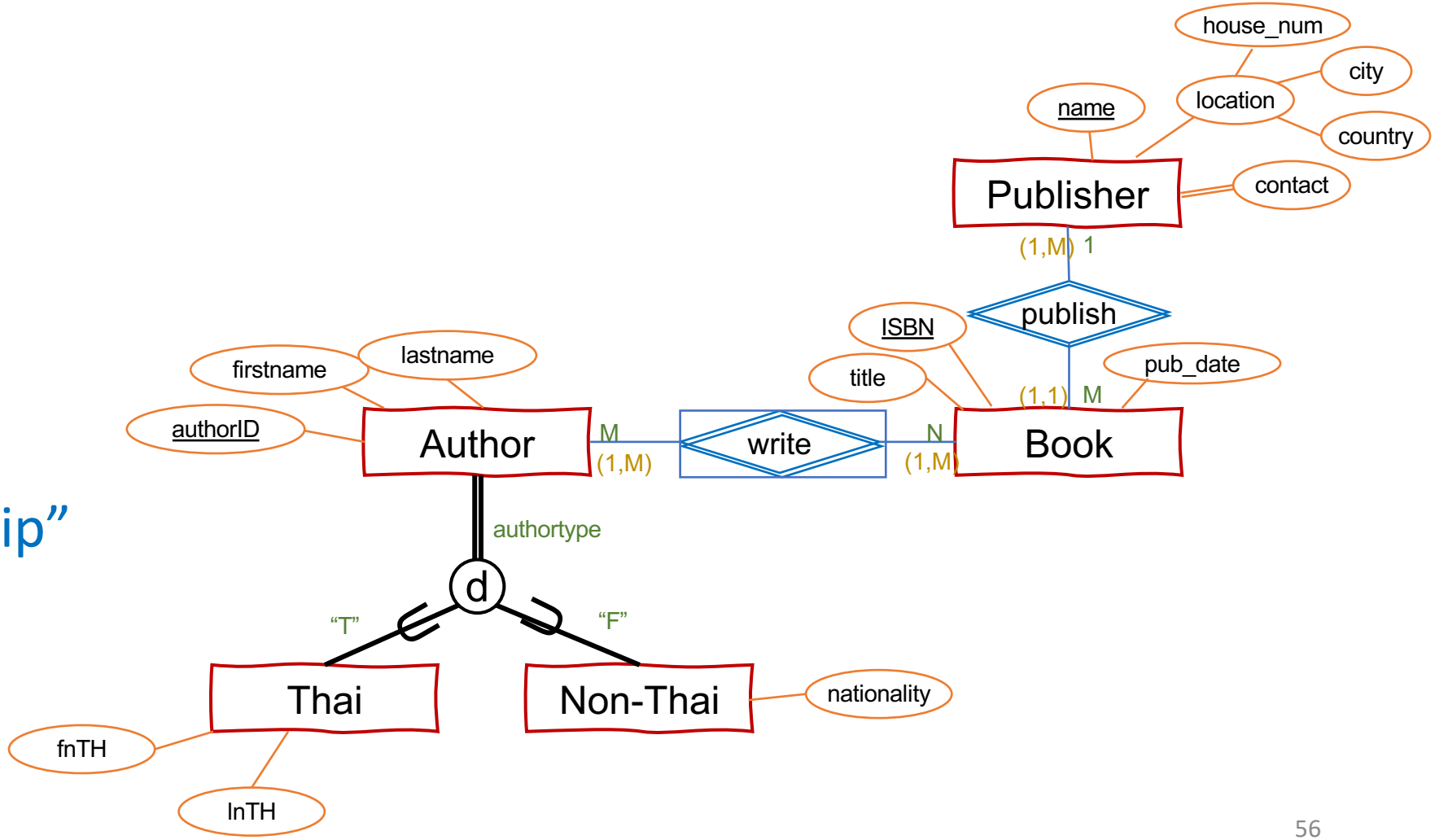
Step 2: Weak Entity

"Not applicable
- no weak entity in the ERD"

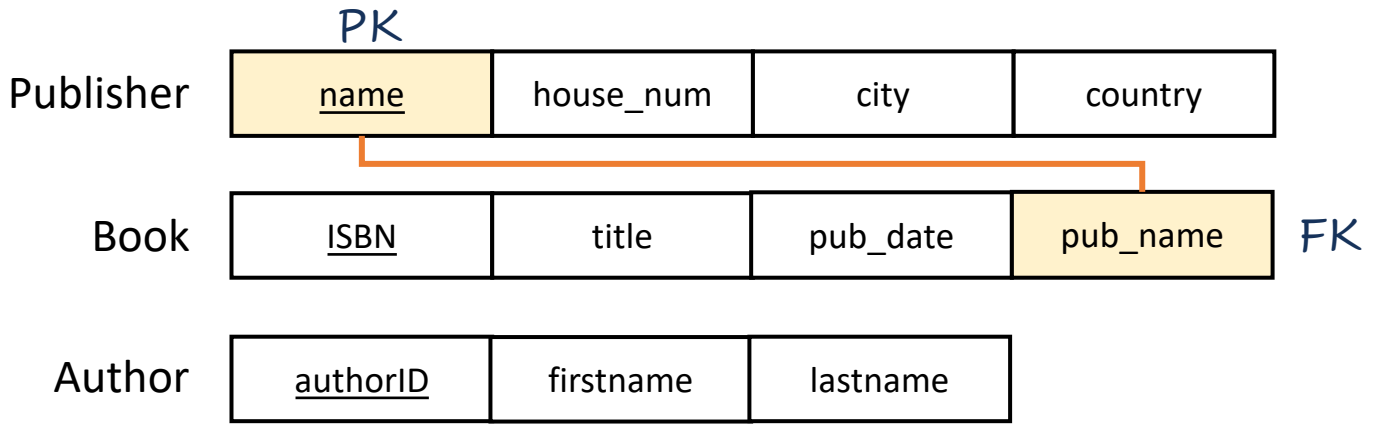


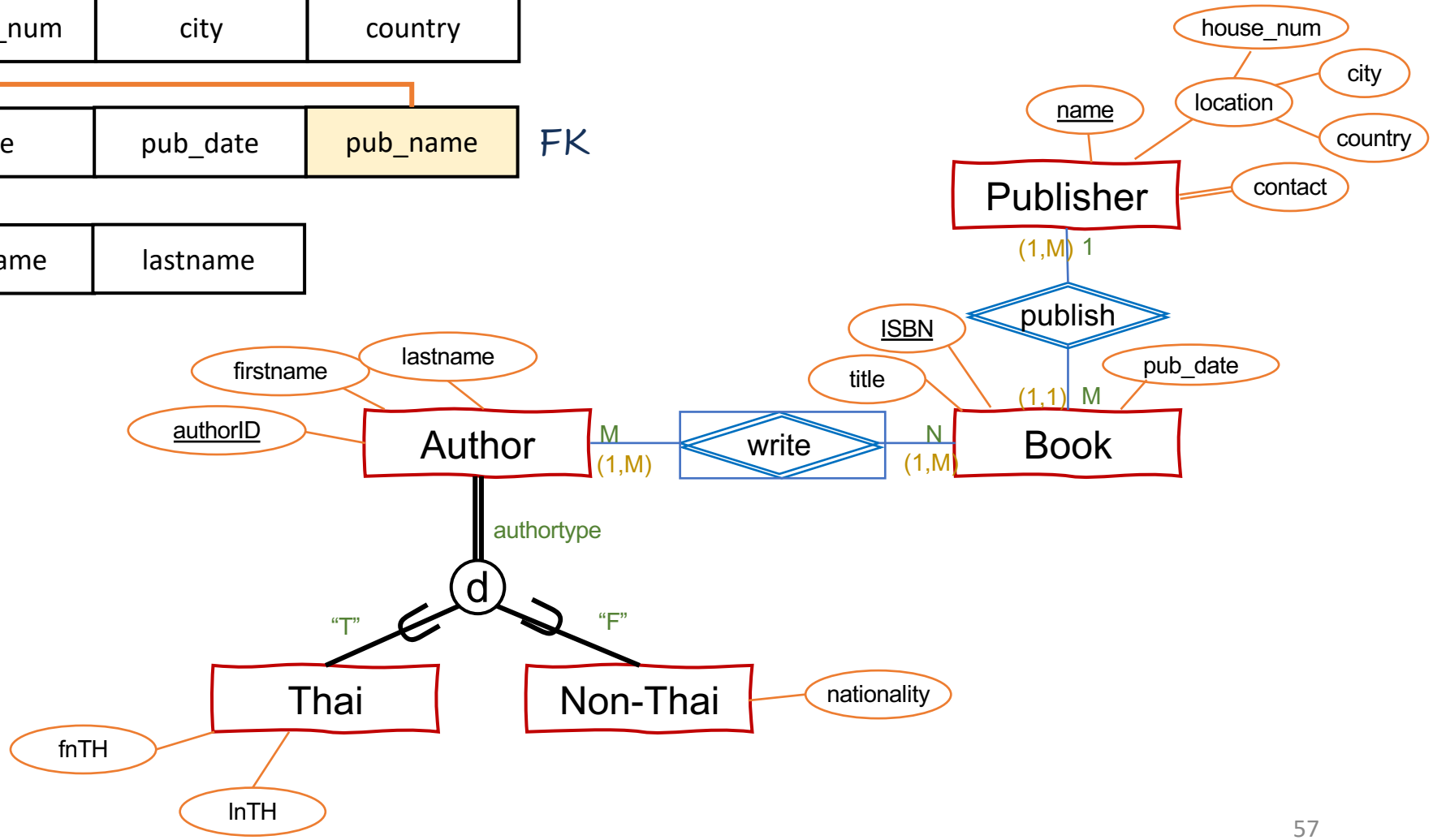
Step 3: 1-1 Relationship

"Not applicable
– no 1-1 Relationship"

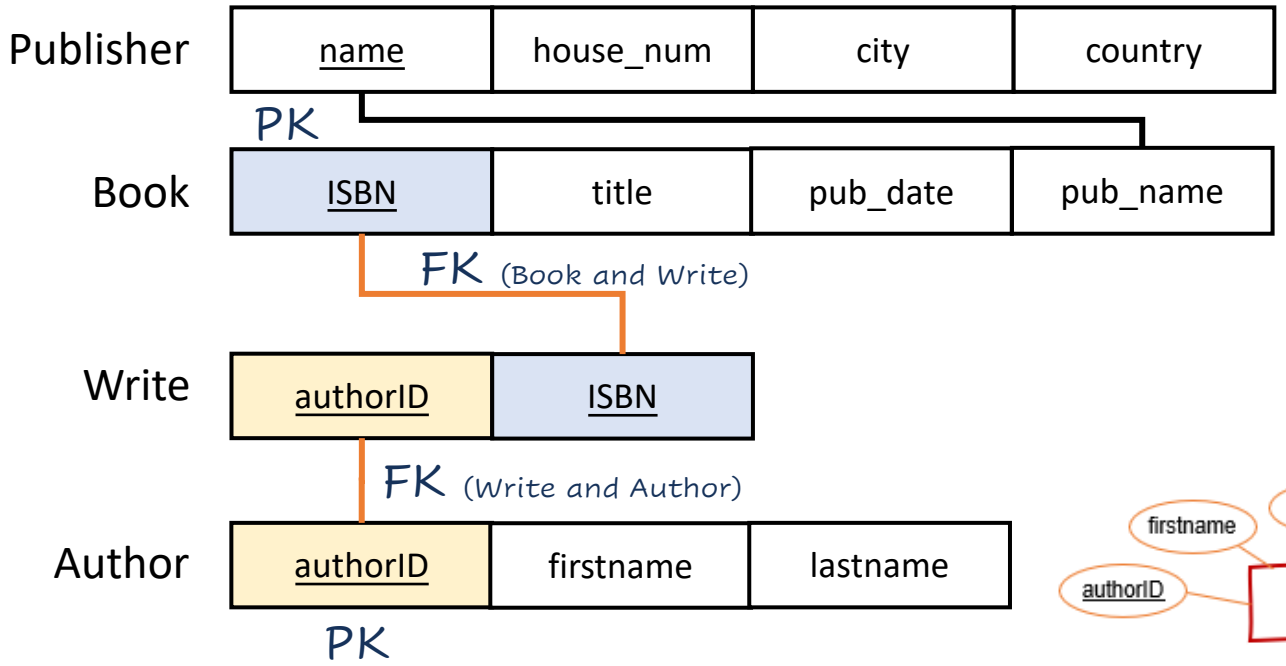


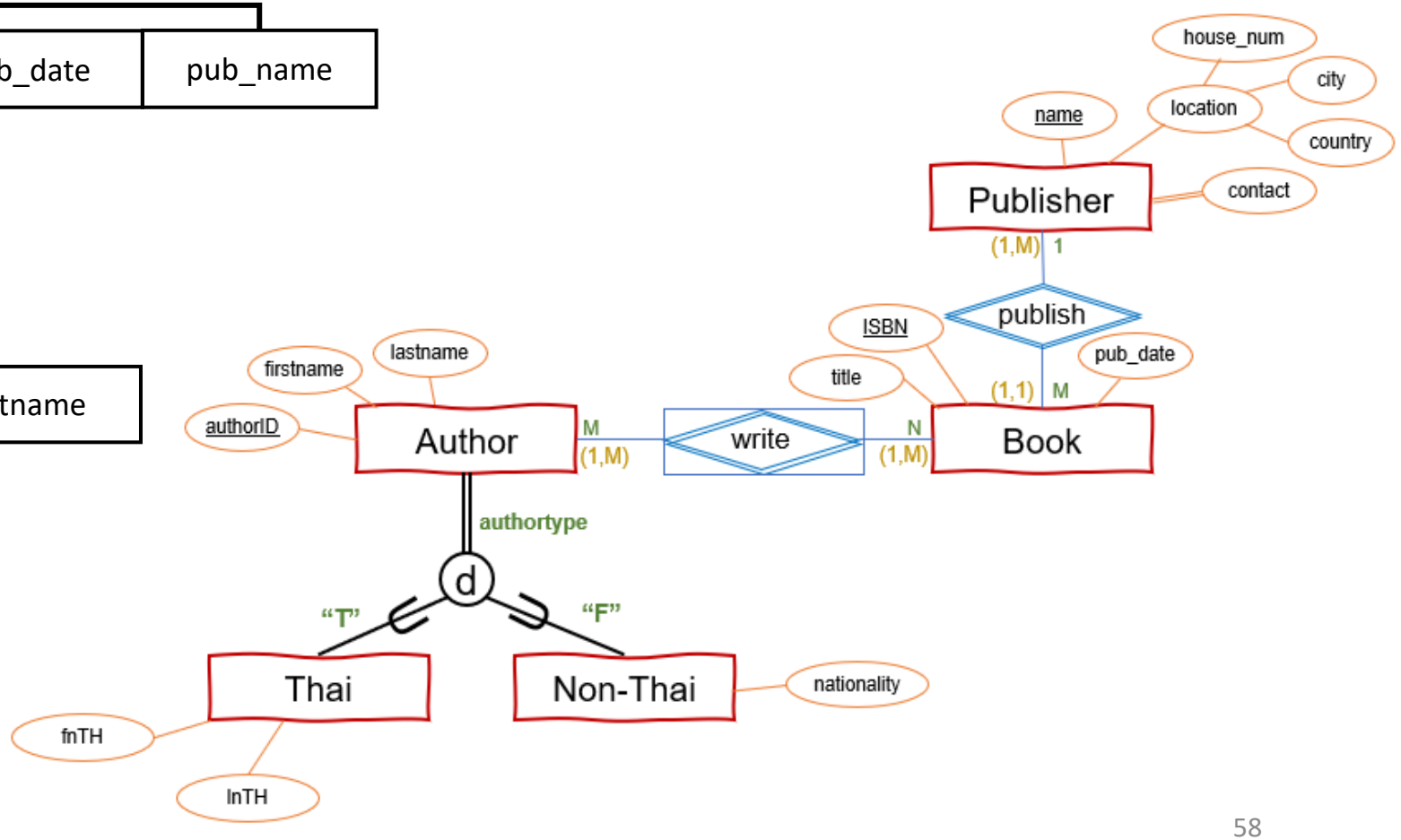
Step 4: 1-M Relationship



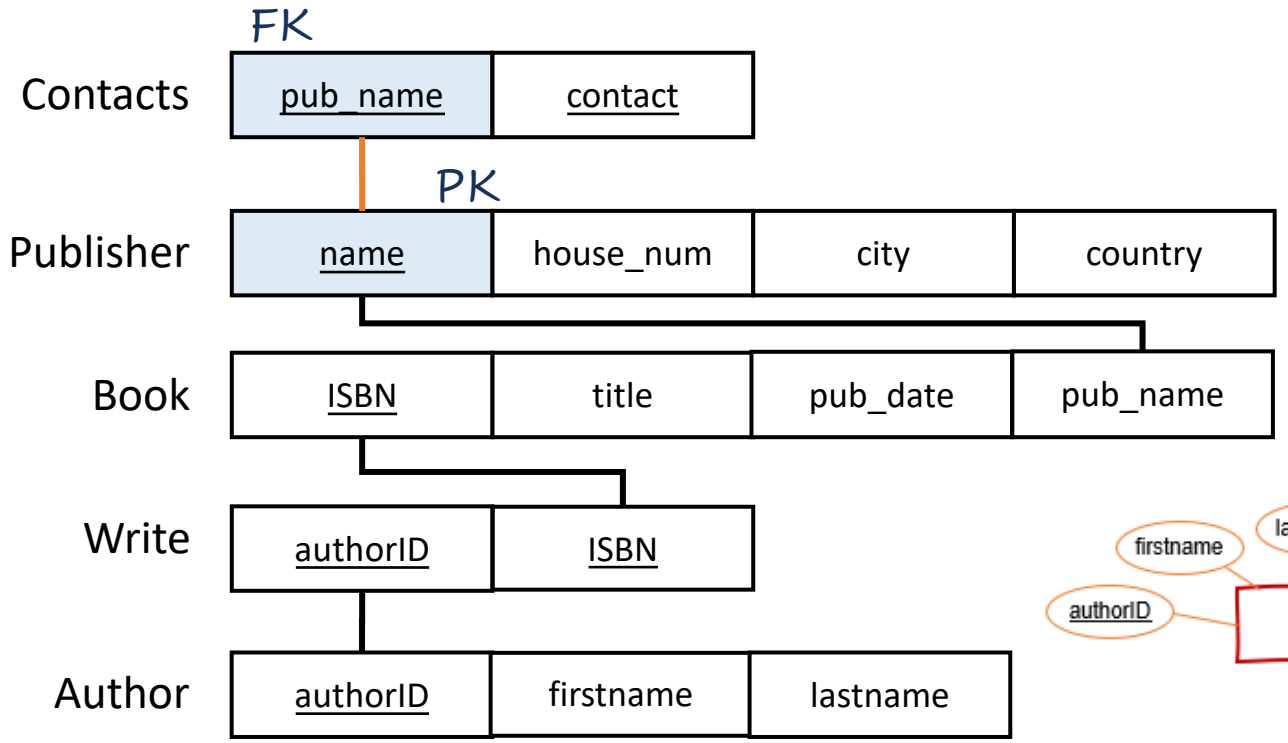


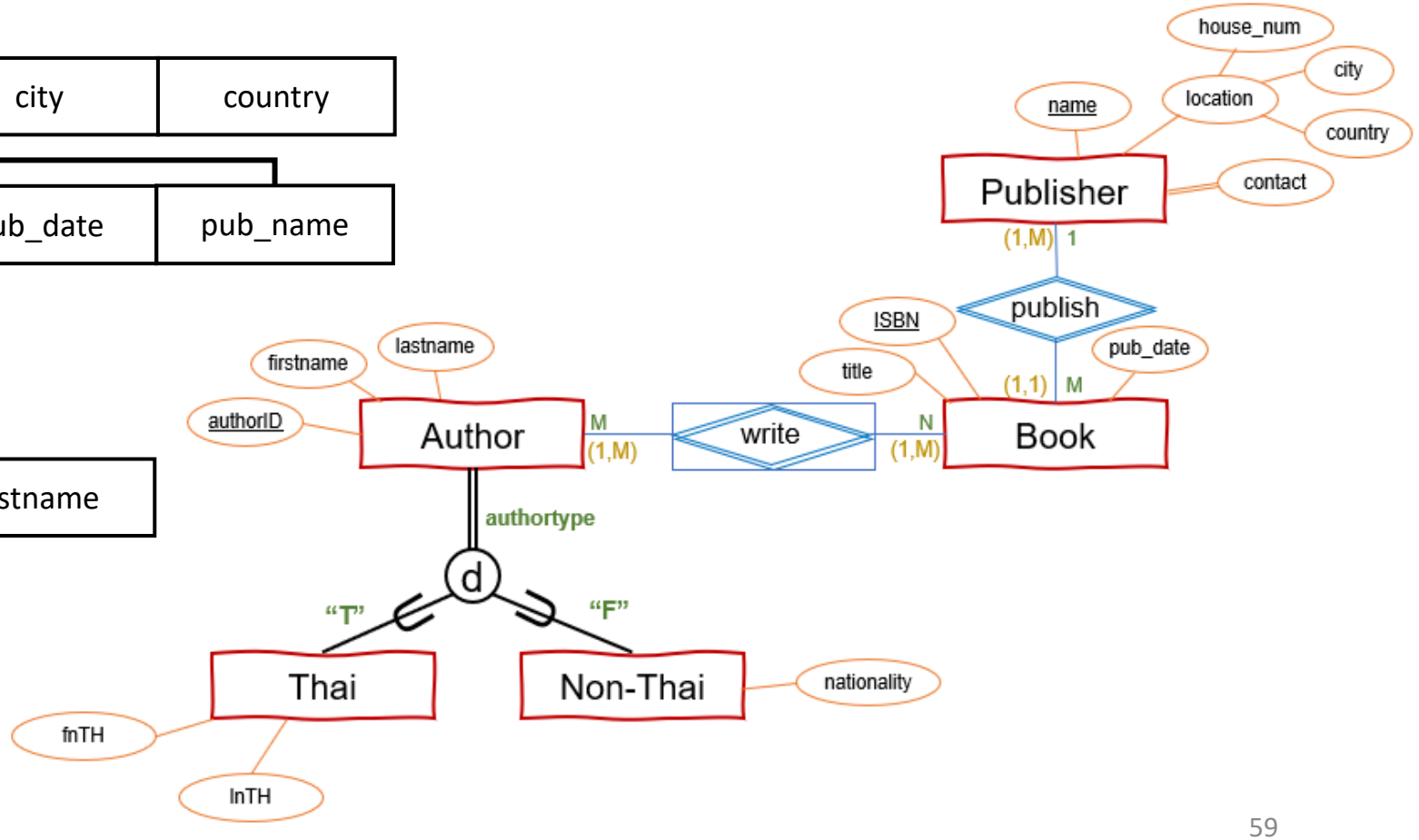
Step 5: M-N Relationship





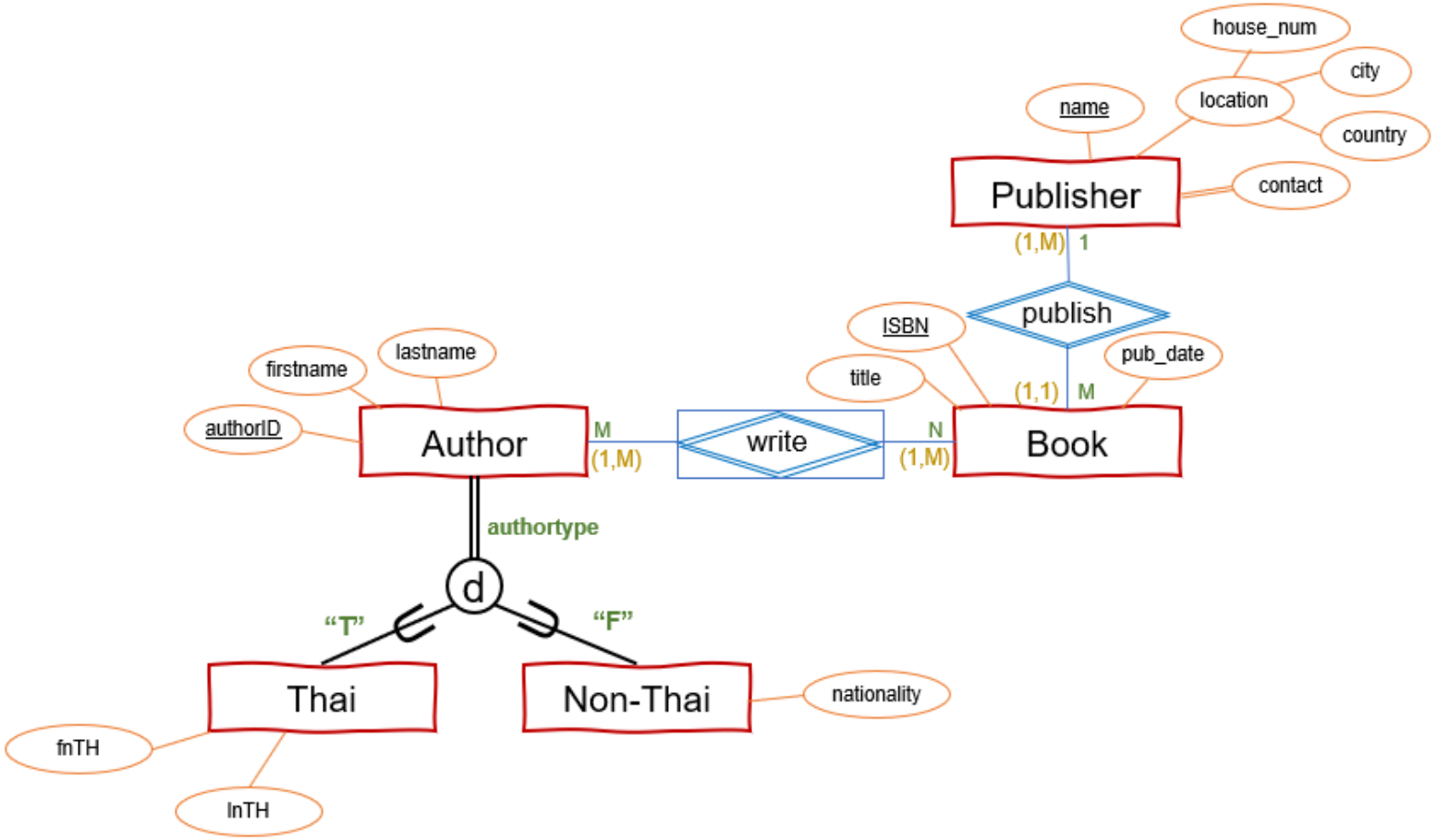
Step 6: Multi-valued Attributes



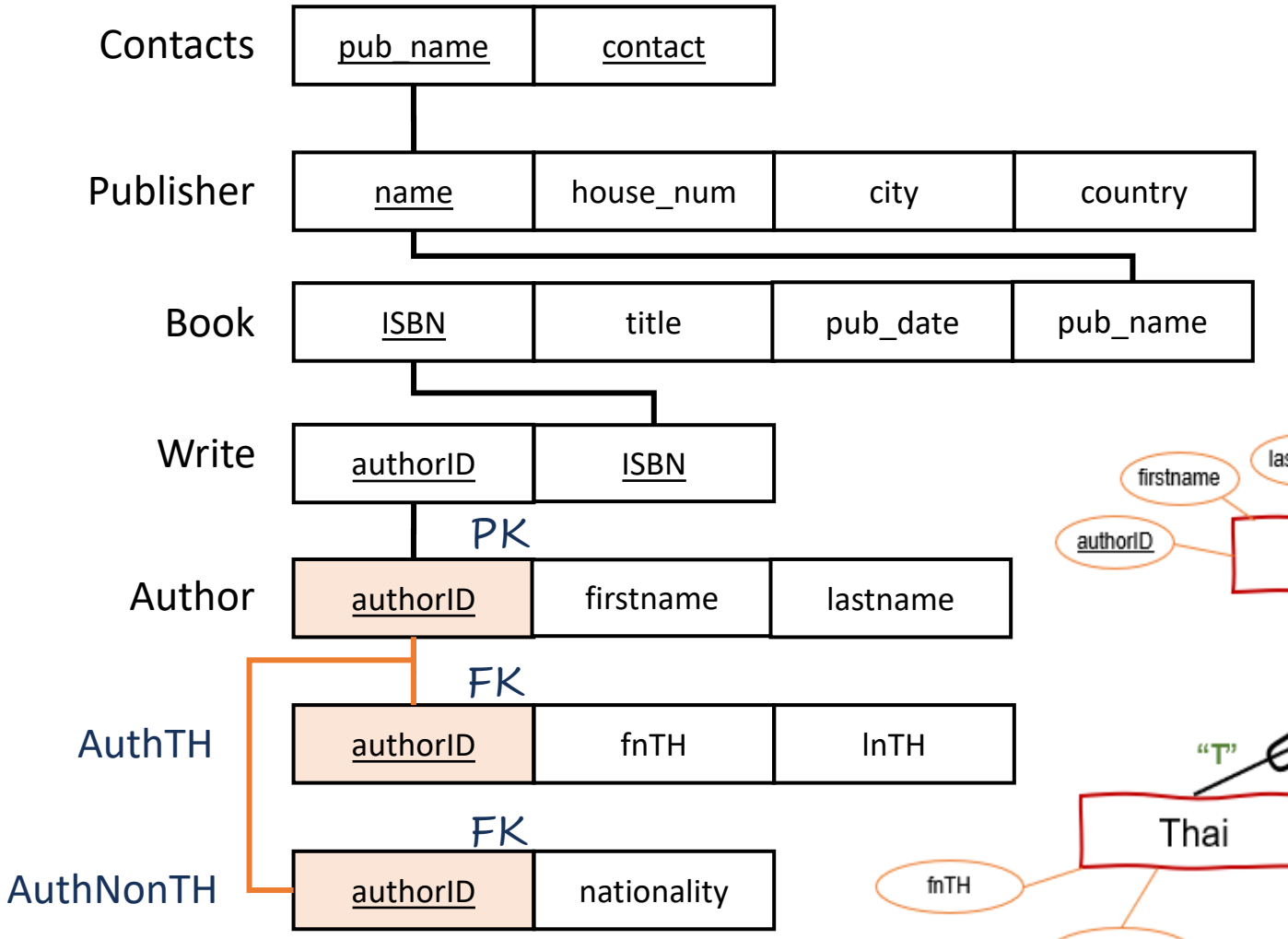


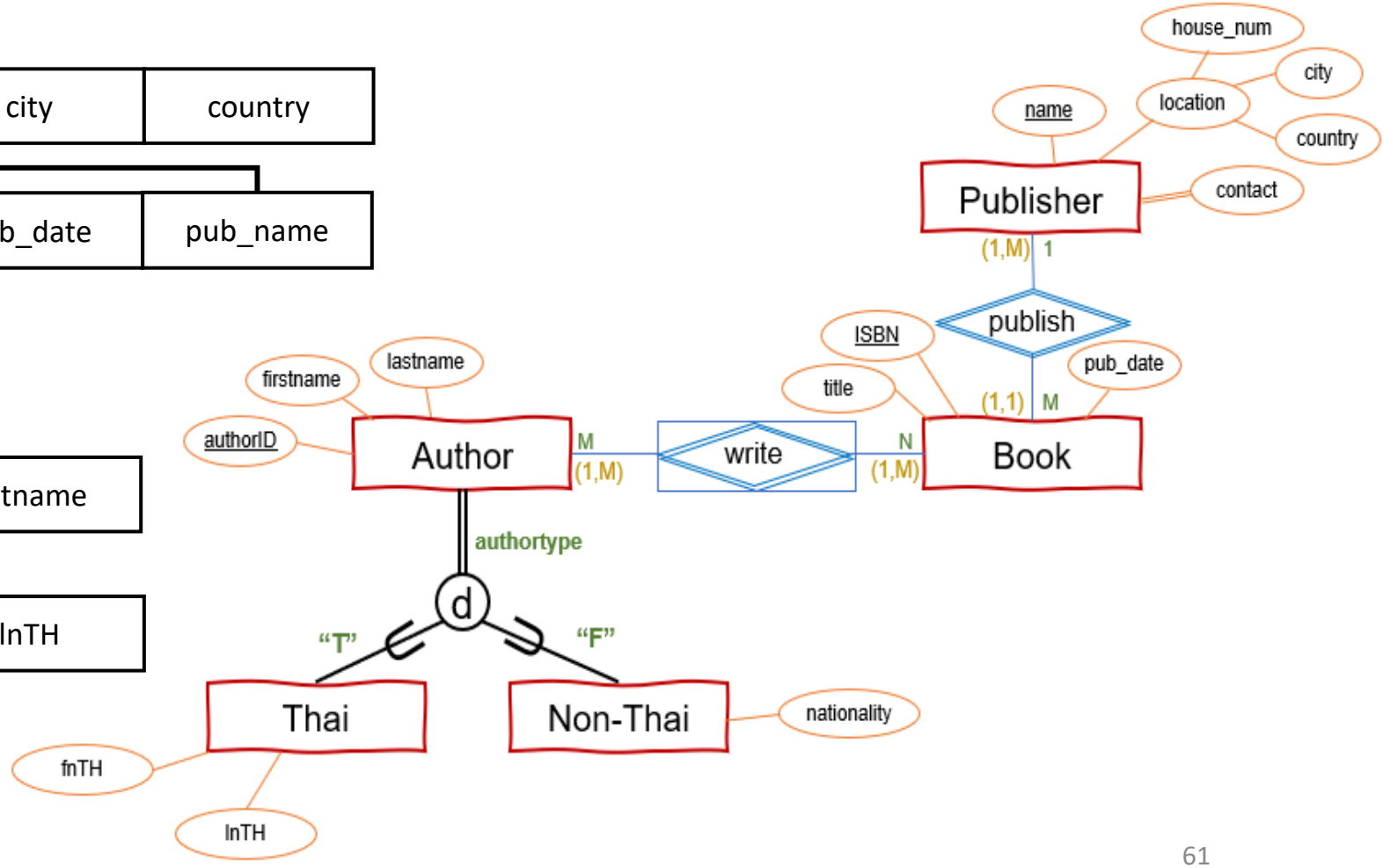
Step 7: N-ary Relationship

"No N-ary Relationship"

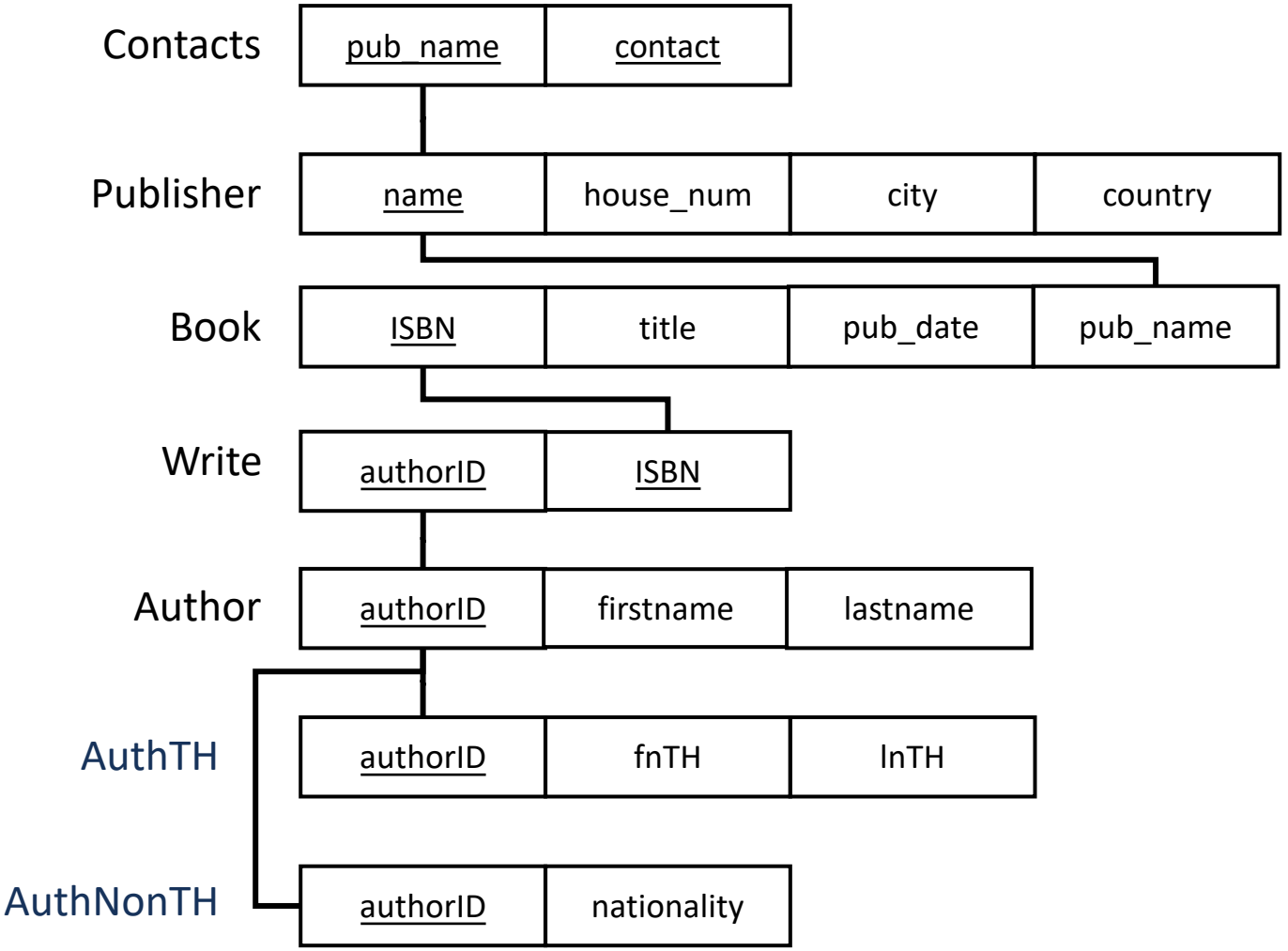


Step 8: Specialization (8A)





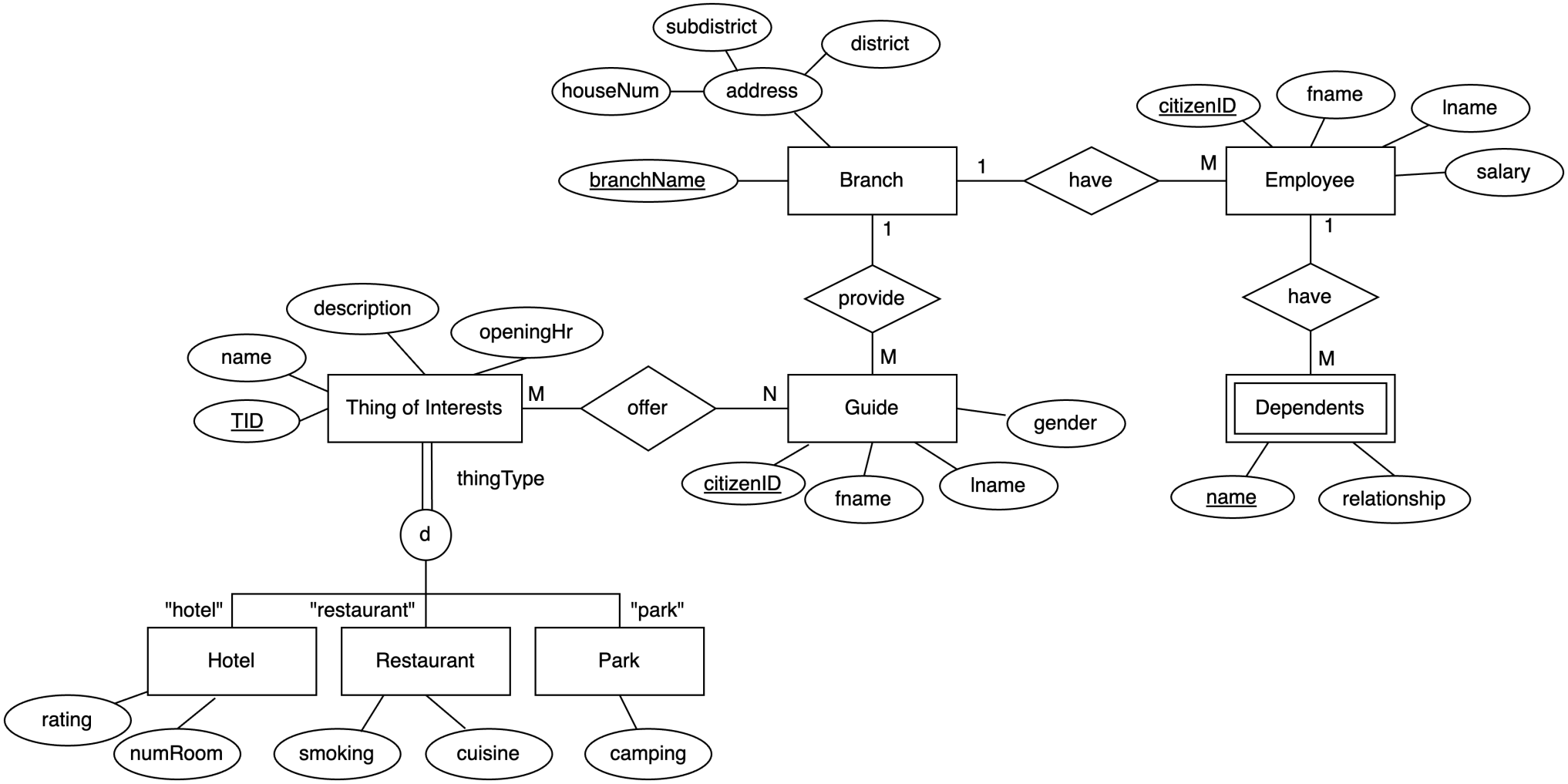
Final Relational Model



In-class Exercise

Thai Travel Agency

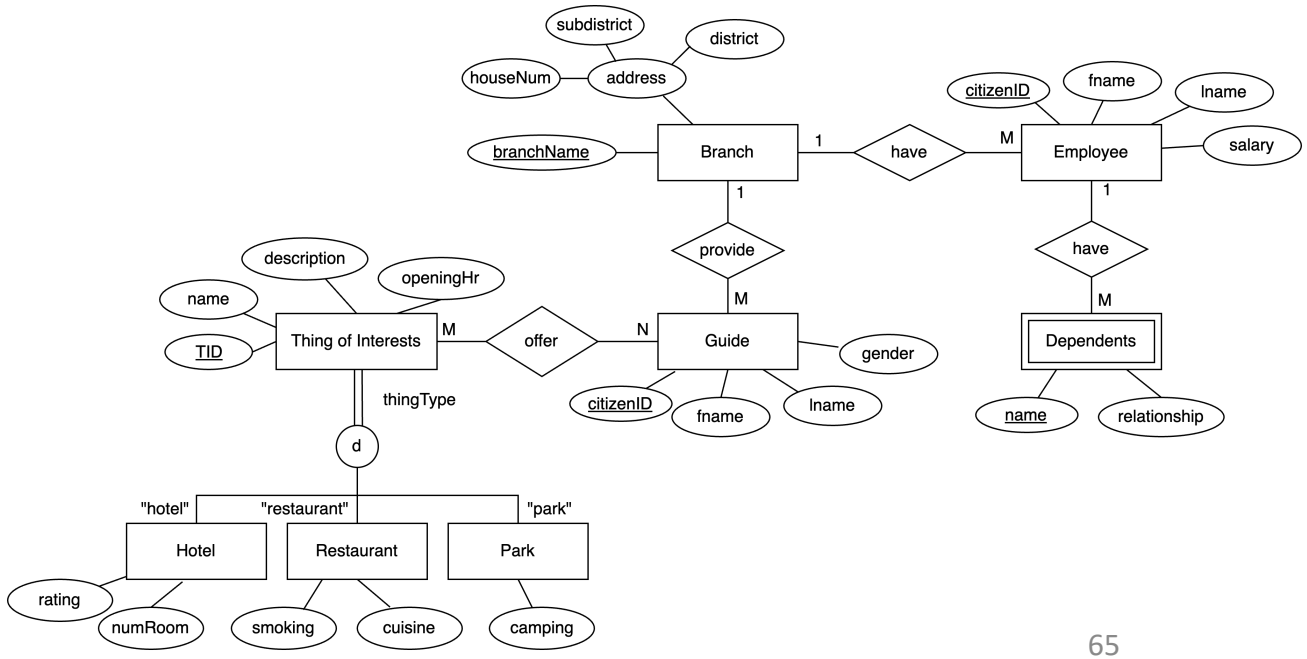
Thai Travel Agency



Step 1: Regualar Entity

address

Branch	branchName	houseNum	subdistrict	district
Employee	ecitizenID	fname	lname	salary
Guide	gcitizenID	fname	lname	gender



Step 2: Weak Entity

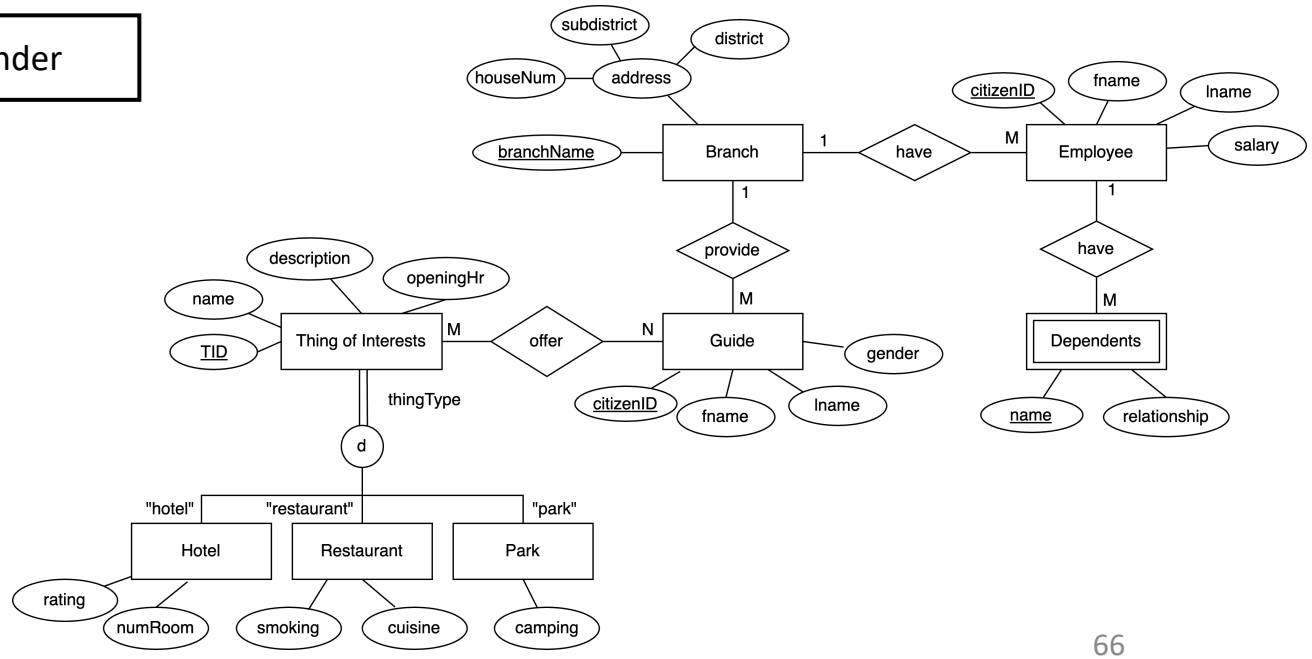
FK PK

Dependents	ecitizenID	name	relationship

Employee	ecitizenID	fname	lname	salary

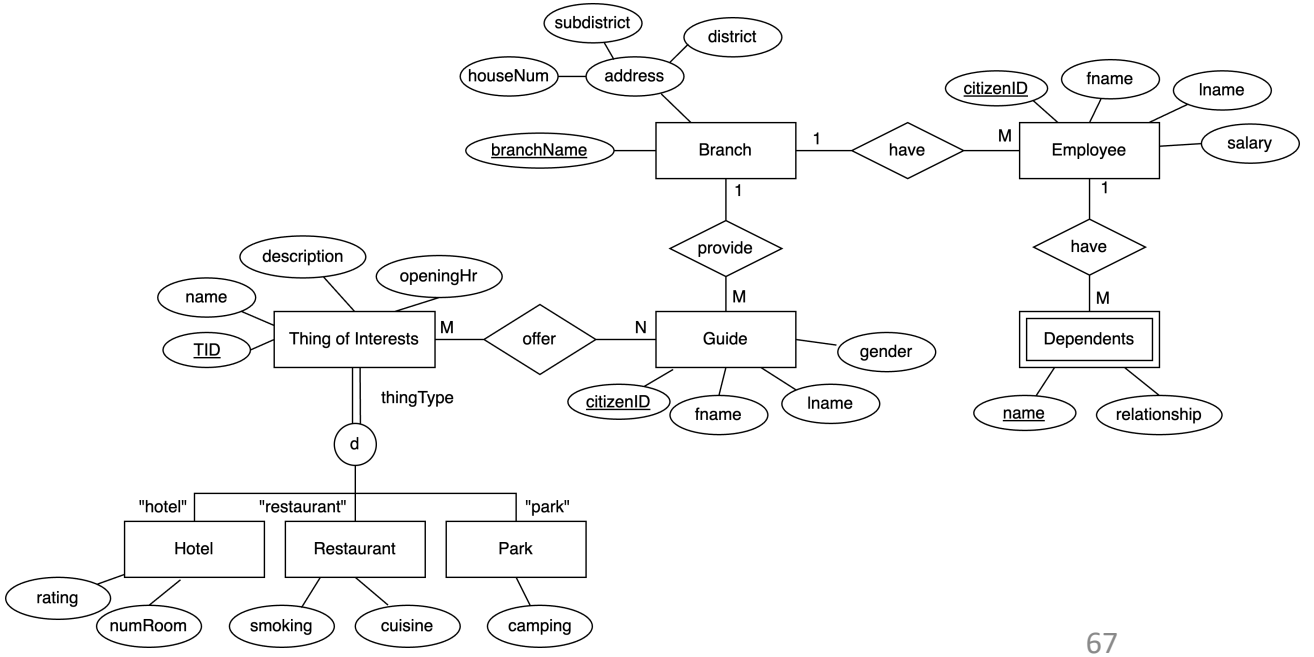
Branch	branchName	houseNum	subdistrict	district

Guide	gcitizenID	fname	lname	gender

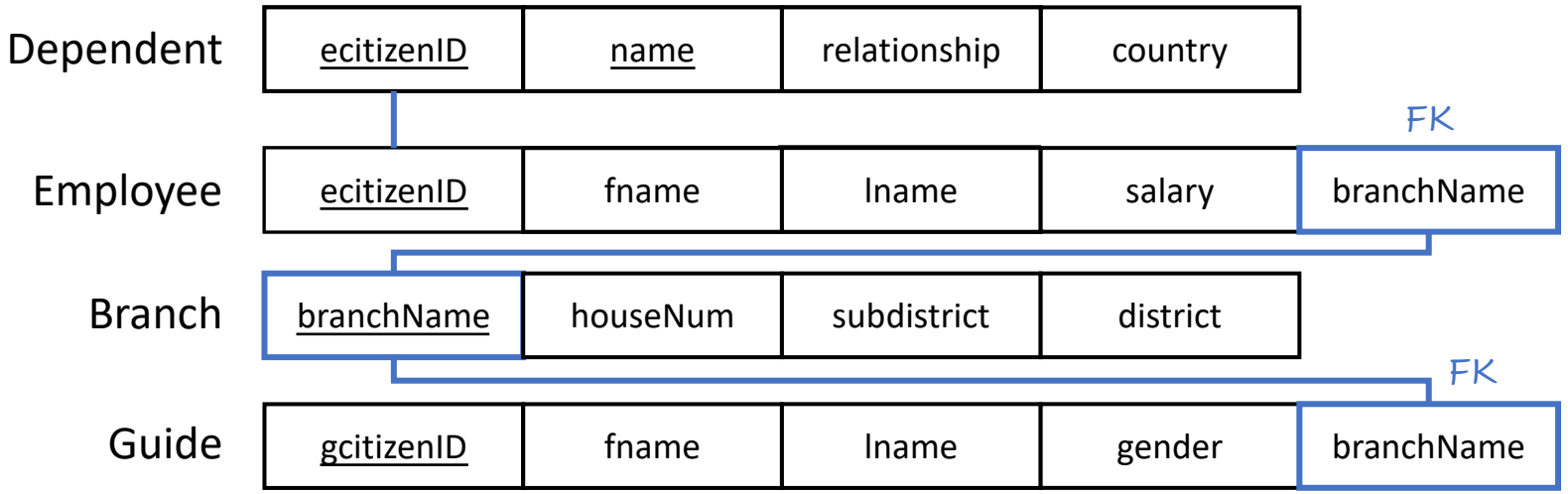


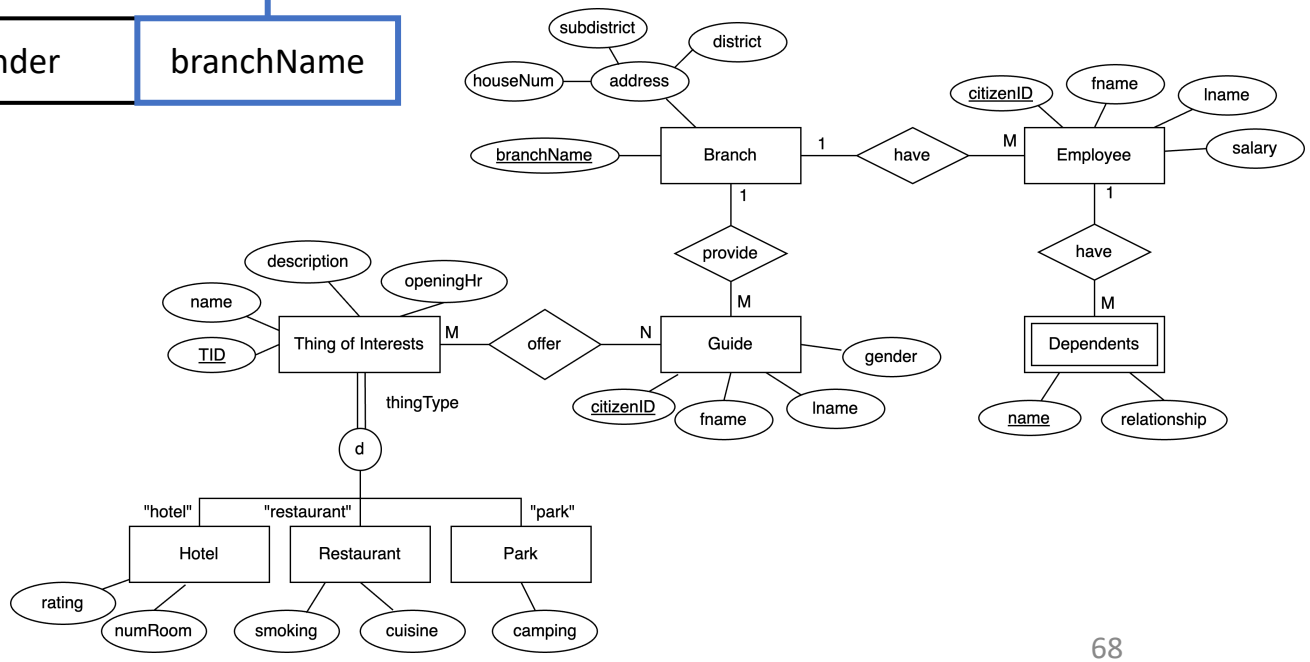
Step 3: 1-1 Relationship

"No 1-1 Relationship"

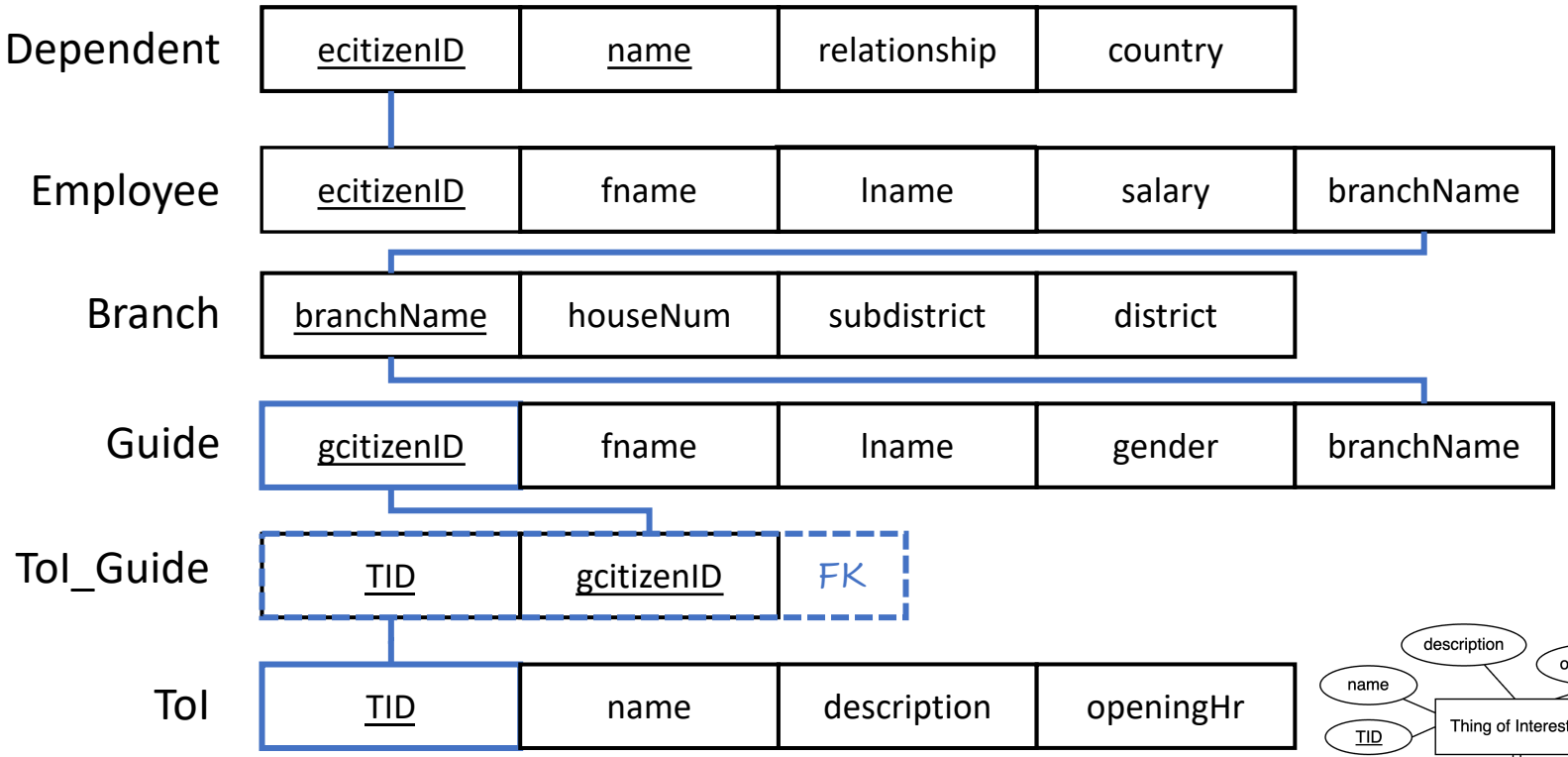


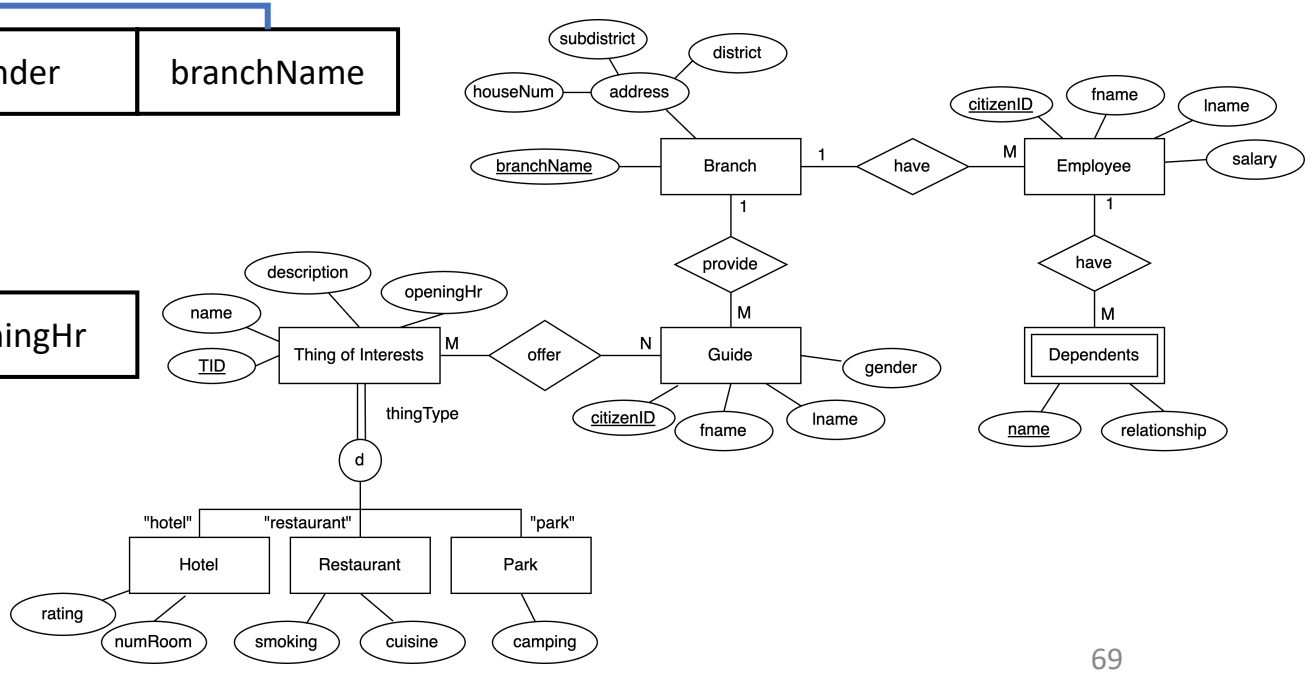
Step 4: 1-M Relationship





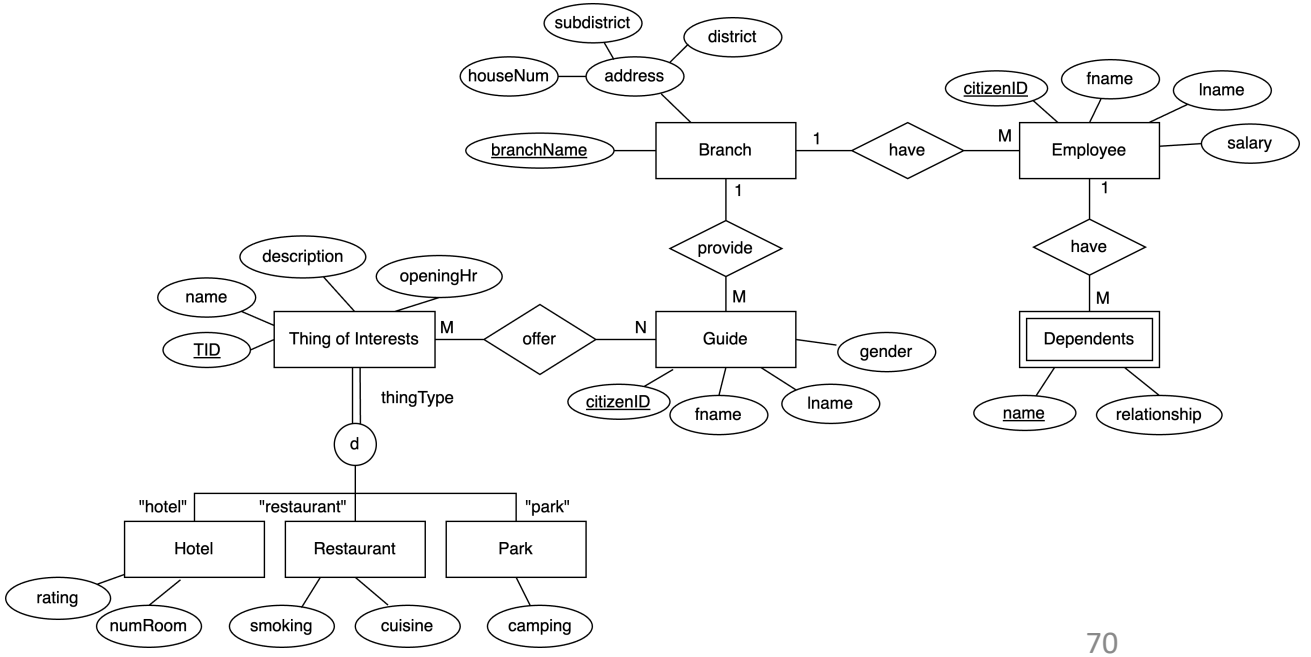
Step 5: M-N Relationship





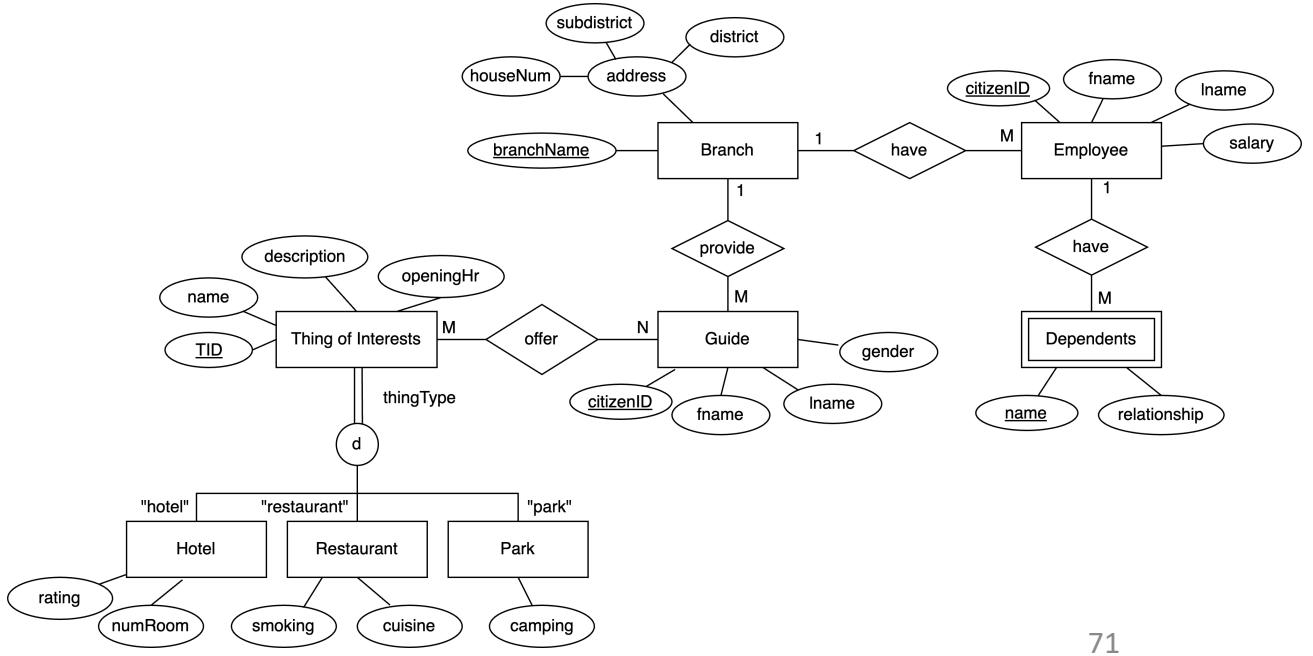
Step 6: Multi-valued Attributes

"No Multi-valued Attributes"

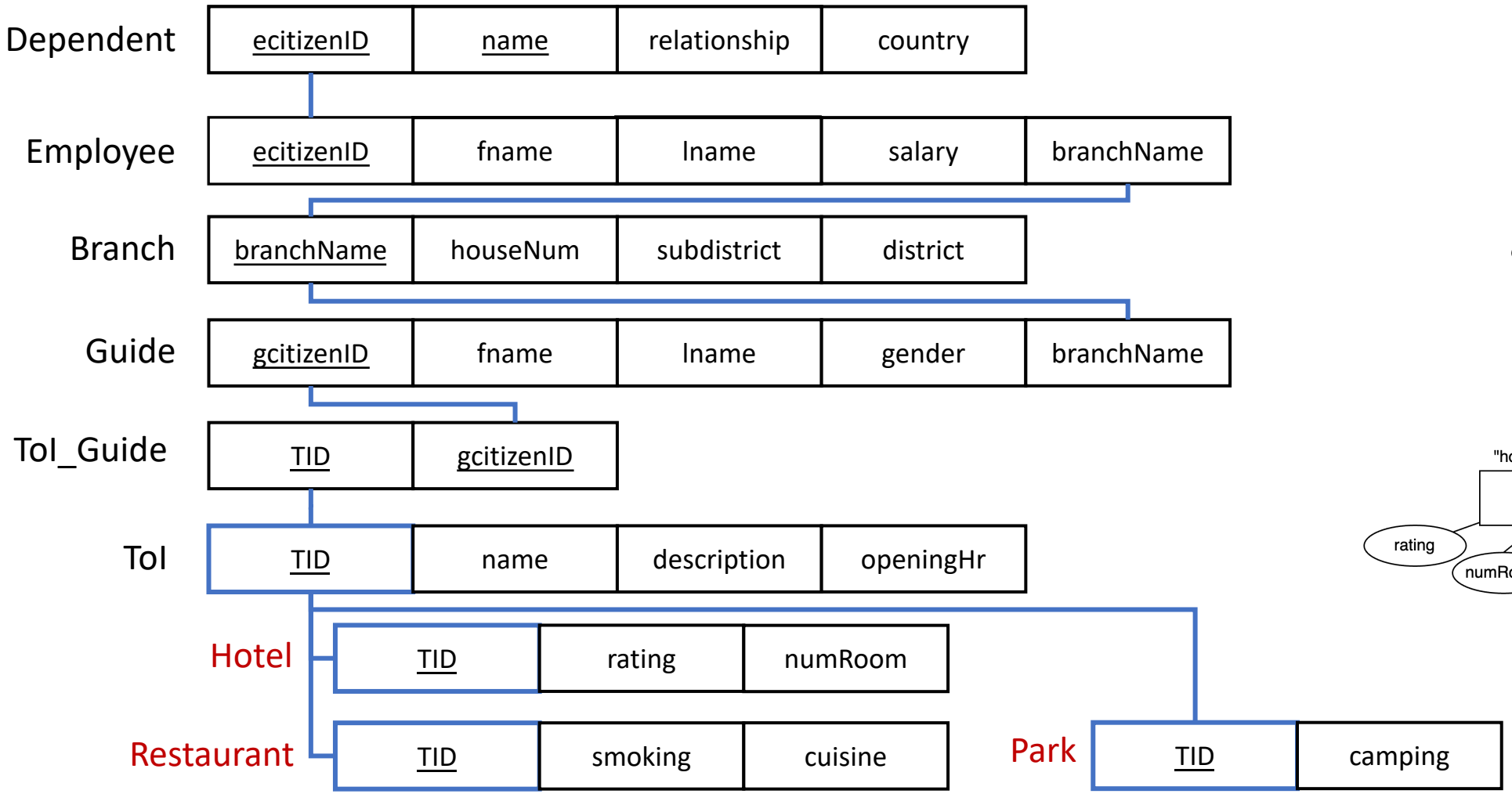


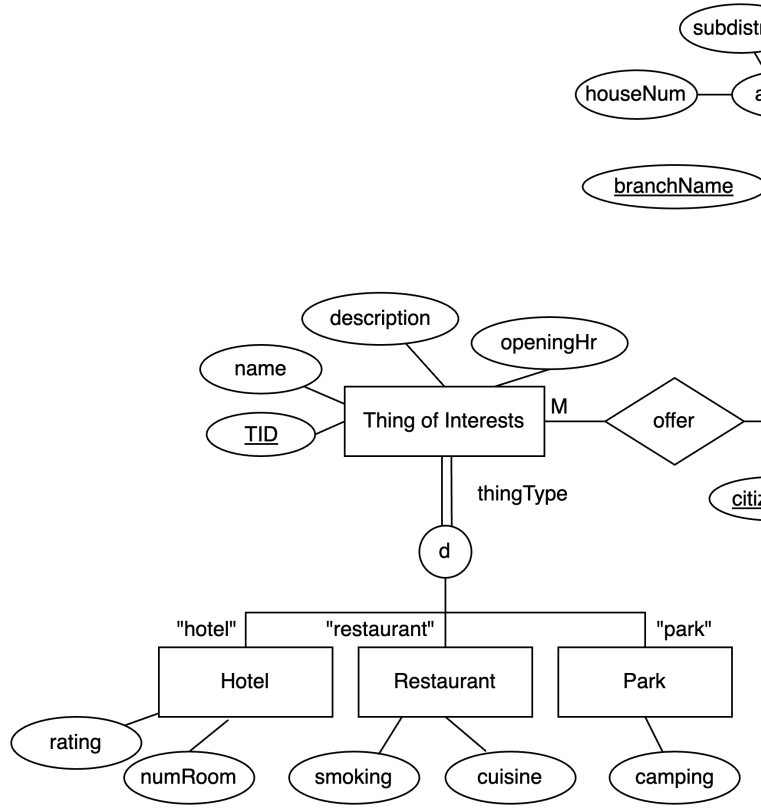
Step 7: N-ary Relationship

"No N-ary Relationship"



Step 8: Specialization (8A)





Final Relational Model

